Expert Systems With Applications 238 (2024) 122334

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

# TAG-DTA: Binding-region-guided strategy to predict drug-target affinity using transformers

Nelson R.C. Monteiro [a,*], José L. Oliveira [b], Joel P. Arrais [a]

[a] Univ Coimbra, Centre for Informatics and Systems of the University of Coimbra, Department of Informatics Engineering, Coimbra, Portugal
[b] IEETA, Department of Electronics, Telecommunications and Informatics, University of Aveiro, Aveiro, Portugal

ARTICLE INFO

*Keywords:*
Drug-target interaction
Binding affinity
Binding pocket
Deep learning
Transformer
Attention

ABSTRACT

The proper assessment of target-specific compound selectivity is paramount in the drug discovery context, promoting the identification of drug-target interactions (DTIs) and the discovery of potential leads. On that account, the accurate prediction of an unbiased drug-target binding affinity (DTA) metric is pivotal to understanding the binding process. Most *in silico* computational approaches, however, neglect the inter-dependency of the proteomics, chemical, and pharmacological spaces and the explainability during the model construction. Furthermore, these methods have yet to actively include information associated with binding pockets during the learning process, which is essential to DTA prediction performance and model explainability. In this study, we propose an end-to-end binding-region-guided Transformer-based architecture that simultaneously predicts the 1D binding pocket and the binding affinity of DTI pairs, where the prediction of the 1D binding pocket guides and conditions the prediction of DTA. This architecture uses 1D raw sequential and structural data to represent the proteins and compounds, respectively, and combines multiple Transformer-Encoder blocks to capture and learn the proteomics, chemical, and pharmacological contexts. The predicted 1D binding pocket conditions the attention mechanism of the Transformer-Encoder used to learn the pharmacological space in order to model the inter-dependency amongst binding-related positions. The results show that the proposed architecture, TAG-DTA, achieved the best performance in DTA prediction compared to state-of-the-art benchmarks, including in unknown subsets of the proteomics and chemical representation spaces. Moreover, the 1D binding pocket prediction increases the discriminative power and robustness of the aggregate representation of the pharmacological space and improves the DTA prediction performance. Overall, this research study validates the applicability of an end-to-end Transformer-based architecture in the context of drug discovery, and that combining computationally different yet contextually related tasks is critical to new findings in the DTI domain. Additionally, it shows that TAG-DTA is capable of providing increasing DTI and prediction understanding due to the nature of the attention blocks and prediction of the 1D binding pocket. The data and source code used in this study are available at: https://github.com/larngroup/TAG-DTA.

## 1. Introduction

The discovery of compounds that selectively bind to relevant and ligandable proteins remains one of the greatest challenges in drug discovery. In recent years, *in silico* approaches have gained acceptance due to their ability to exploit comprehensive chemical and proteomic libraries, overcome the drawbacks of high-throughput bioassays, and improve the efficacy of the early stages of drug discovery (Schneider et al., 2020). For this reason, numerous computational methods have been explored to solve the challenge of drug-target interaction (DTI) prediction, focusing on binary classification during the inference process (D'Souza et al., 2020). Although these methods can be classified into different domains, including docking simulation (Wang et al., 2019; Zhang et al., 2020) or quantitative structure–activity relationship (QSAR) modeling (Luo et al., 2014; Ma et al., 2015), the increasing amount of biological and chemical data has led to predictive solutions based on proteochemometric (PCM) approaches (Bongers et al., 2019). PCM studies explore various properties and representations to characterize proteins and compounds, including Morgan or extended-connected fingerprints (Morgan, 1965; Rogers & Hahn, 2010), evolutionarily conserved profiles (Altschul et al., 1997), CDK (Willighagen et al., 2017) descriptors, physicochemical properties (van Westen, Swier, Cortes-Ciriano et al., 2013; van Westen, Swier, Wegner et al., 2013), or PaDEL-Descriptor (Yap, 2011) features. These explicit ligand-derived and target-derived descriptors are typically combined

* Corresponding author.
*E-mail addresses:* nelsonrcm@dei.uc.pt (N.R.C. Monteiro), jlo@ua.pt (J.L. Oliveira), jpa@dei.uc.pt (J.P. Arrais).

https://doi.org/10.1016/j.eswa.2023.122334
Received 12 July 2023; Received in revised form 21 October 2023; Accepted 23 October 2023
Available online 26 October 2023

and used as input to machine learning models and deep learning architectures, including random forest (RF) (Coelho et al., 2016), support vector machine (SVM) (Peng et al., 2017), feed-forward neural networks (FFN) (Tian et al., 2016), or long short-term memory (LSTM) neural networks (Wang et al., 2020). To overcome the limitations of using global descriptors, recent PCM studies have explored the use of 1D structures, such as amino acid sequences and SMILES (Simplified Molecular Input Line Entry System) strings, and graph representations in combination with convolutional neural networks (CNNs) and graph neural networks (GNNs) (Lee et al., 2019; Monteiro et al., 2021; Tsubaki et al., 2019). Despite the interesting findings and encouraging results obtained in DTI classification studies, the use of shallow binary associations to characterize DTI pairs limits the quality of the results, leading to a lack of target selectivity.

The increase in interactions with known bioactivity measurements and the expansion of binding-related databases such as ChEMBL (Gaulton et al., 2017), BindingDB (Gilson et al., 2016), or PDBbind (Liu et al., 2017) have been instrumental in the pursuit of more realistic and informative studies, i.e., drug-target affinity (DTA) prediction. The DTA prediction problem is regarded as a regression task and it is substantially more challenging than binary DTI classification. It has initially been driven by the limitations of the scoring functions used in virtual screening. On that account, machine learning models and deep learning architectures combined with 3D structural features (Kumar & Kim, 2021; Meli et al., 2021) or 3D single instance learning (Jiménez et al., 2018; Jones et al., 2021) have been proposed as potential replacements for these scoring functions. Recent studies, however, have been exploring PCM approaches based on chemogenomic and lower structural information to conduct their experiments, overcoming the limitations of the 3D structure space and leading to more reproducible methodologies in the DTA prediction domain. In addition to some traditional machine learning algorithms, including Kronecker-regularized least squares (Pahikkala et al., 2014) or gradient boosting regression trees (He et al., 2017), many studies explore CNN-based models, such as 1D CNNs, 2D CNNs, or graph CNNs (GCNs) to extract knowledge and meaningful information from different protein and compound representations for the prediction of binding affinity (Abbasi et al., 2020; Nguyen et al., 2020; Öztürk et al., 2018; Rifaioglu et al., 2021; Shim et al., 2021; Wang et al., 2021). Most studies, especially those based on deep learning, however, focus on sources of unreliable binding affinity metrics such as inhibition constant ($K_i$), half maximum inhibition concentration ($pIC_{50}$), or KIBA (Tang et al., 2014) scores to increase the performance of their models due to the number of available data points. Even though deep learning-based architectures perform significantly better when the dataset becomes large, the use of biased bioactivity metrics limits the validity of the results in the pharmacological domain.

Given the advances in computational power, most recent studies dealing with DTI or DTA prediction have explored deep learning strategies, achieving better results than traditional machine learning solutions (Rifaioglu et al., 2018). Despite the increasing modular ability of these architectures to learn sequential and/or structural motifs and extract robust representations, the final predictions are mostly not interpretable by humans, which affects the understanding of the underlying aspects around the inner decisions (Castelvecchi, 2016; London, 2019). Moreover, several of these models neglect the interdependency of the sequential and structural units of each binding component (and their intra-associations) or the inter-associations that revolve around the binding substructures (context of interaction), leading to predictions based on local and independent (without context) scattered motifs (Agamah et al., 2019; Hanson et al., 2020; Schenone et al., 2013). In order to overcome the aforementioned limitations, attention-based models have been proposed to learn short and long-term context dependencies amongst the units of proteins or compounds and condition the weight given to input elements. For instance, Zhao et al. (2022) proposed HyperAttentionDTI, where 1D CNNs are used to identify local patterns and extract features from protein sequences and SMILES strings, and a sigmoid-based attention mechanism to model the inter-associations between the units of each binding component. Inspired by the remarkable success of Transformers (Vaswani et al., 2017) in several computational domains and their potential to capture features between two sequences, Chen et al. (2020) proposed TransformerCPI, which employs a classification encoder–decoder scheme to predict DTIs. In their approach, protein sequences are converted into sequential representations and used as input to the encoder, which is based on gated 1D CNNs. On the other hand, SMILES strings are converted into graph representations and propagated through a GCN to obtain atomic features. These molecular features are used as input to the Transformer-Decoder, which learns the interaction sequence. Motivated by the effectiveness of stacking Transformer-Encoders to extract an augmented contextual representation of the input (Devlin et al., 2019), Huang et al. (2021) introduced an architecture based on Transformer-Encoders and CNNs, where the Transformer-Encoders are used in the feature extraction process to learn the intra-associations amongst 1D substructures in proteins or compounds, and the CNN to model the higher-order interaction. To learn the joint contribution of the individual units of each binding component and the involving interaction substructures for the prediction of DTA, Monteiro, Oliveira et al. (2022) proposed DTITR, which stacks self-attention and cross-attention Transformer-Encoders into an end-to-end framework.

In spite of the superior performance of attention-based architectures and the ability to provide increased model understanding in the DTI domain, the reliability of the predictions is limited considering that information about binding sites/pockets is not actively integrated into the learning process. The identification of protein-ligand binding pockets is crucial for understanding the biological functions of proteins and the mechanisms involved in DTIs. On that account, several computational solutions for predicting binding pockets have been proposed. These methods are categorized by the strategy of the algorithm, e.g., geometric, template, or learning-based (machine learning), or by the level of structural data, i.e., sequence or 3D structure-based. For instance, the work of Yang et al. (2013b), COACH, focuses on a template and 3D structure-based representative consensus approach that combines the top-scoring predictions of different algorithms, including TM-SITE (Yang et al., 2013b), S-SITE (Yang et al., 2013b), COFACTOR (Roy et al., 2012), FINDSITE (Brylinski & Skolnick, 2007), and ConCavity (Capra et al., 2009), using a linear SVM. P2Rank (Krivák & Hoksza, 2018) adopts a 3D structure-based approach to predict binding pockets, where residues with high ligandability scores located within a solvent-accessible surface of the protein are clustered to form the binding pocket. The ligandability score is determined using an RF classifier and different features related to the local geometric neighborhood. Motivated by the existing similarities between neural machine translation and binding residue prediction, Cui et al. (2019) proposed a sequence-based approach, DeepCSeqSite, in which CNNs are used to identify and extract motifs from protein sequences, and each residue is predicted as a binding or non-binding residue.

Based on these reported characteristics and drawbacks, we propose an end-to-end Transformer-based framework to simultaneously predict the 1D binding pocket and the DTA measured in terms of the dissociation constant ($K_d$), where the prediction of the binary binding vector, which states the binding nature of each protein residue, guides (conditions) the prediction of the binding affinity. The targets and compounds are represented using 1D sequential and structural information, specifically protein sequences and SMILES strings, respectively. This architecture, TAG-DTA, consists of two Transformer-Encoder-based prediction models, specifically a 1D binding pocket classifier and a binding affinity regressor, and shares three cores layers, including lower Transformer-Encoders and a condition-based concatenation block. We leverage the use of lower self-attention layers to learn the short and long-term proteomics and chemical context dependencies between the sequential and structural units of the proteins and compounds, respectively, and a condition-based concatenation layer to represent the pharmacological (interaction) space. The binding pocket Transformer-Encoder uses the pharmacological space representation for binary token labeling, where the predicted binary 1D binding pocket is used to condition the attention mechanism of the binding affinity Transformer-Encoder, resulting in the exchange of information between the proteomics and chemical domains over binding-related residues. The resulting aggregate representations of the proteomics, chemical, and binding-region-based pharmacological spaces are concatenated and fed into a fully-connected feed-forward network (FCNN), which predicts the binding strength of DTI pairs. The experimental results show that the proposed model can achieve superior binding affinity prediction performance in comparison to state-of-the-art benchmarks, including in experimental settings associated with novel proteins, compounds, and DTI pairs. Furthermore, it provides increasing DTI and model understanding not only due to the nature of the attention blocks, which give information about the overall importance of the input components and their intra-associations, but also due to the prediction of the 1D binding pocket, which determines the attention mechanism over the interaction space and shows explicit evidence of potential key residues within the protein sequences for the binding process.

## 2. Material and methods

### 2.1. Binding affinity dataset

To establish the binding affinity prediction model, we collected drug-target pairs from the Davis et al. (2011) research study, which comprises selectivity assays associated with the human catalytic protein kinome measured in terms of a quantitative dissociation constant ($K_d$). This study covers the interaction between 442 kinases and 72 kinase inhibitors, resulting in 31 824 DTIs. $K_d$ expresses a direct and unbiased measurement of the equilibrium between the receptor–ligand complex and the dissociation components, where lower values are associated with strong interactions.

The protein sequences of the Davis dataset were collected from UniProt (Consortium, 2023) using the corresponding accession numbers. Proteins are characterized by a unique amino acid sequence, resulting in varying sequence lengths. To standardize the number of features and avoid the loss of relevant sequential information or increasing noise, we have fixed the sequence length between 264 and 1400 residues based on a 95% information density threshold. Considering the computational complexity of the self-attention layers of $O(n^2)$ concerning the sequence length, we have applied the approach proposed in the Huang et al. (2021) research study, which combines a frequent consecutive subsequence (FCS) mining method with the byte pair encoding (BPE) algorithm, to represent and encode the protein sequences. This method decomposes each protein sequence into an ordered set of non-overlapping frequent subsequences, where the aggregation of all subsequences must recover the original sequence. The hierarchy dictionary of frequent subsequences ($V_P$) comprises 16 693 different subwords.

The SMILES strings of the Davis dataset were extracted from PubChem (Kim et al., 2023) based on their PubChem CIDs. To ensure a consistent notation to represent the chemical structure of all compounds, we have applied the RDKit (Landrum, 2021) canonical transformation to every SMILES string. Similarly to the protein sequences, we have selected only SMILES strings with a sequential length between 38 and 72 chemical characters. To represent and encode the SMILES strings into numerical values, we have used an integer-based encoding based on a character-integer dictionary, which converts each chemical token into the corresponding numerical value. The character-integer dictionary ($V_S$) contains a total of 72 unique letters (labels) resulting from the scanning of approximately 1.3 M SMILES strings from the ChEMBL (Gaulton et al., 2017) database.

The distribution of the Davis $K_d$ values is significantly skewed toward $K_d$ equal to 10 000 nM, which is associated with extremely weak or almost non-existing interactions. Moreover, the variance of the distribution is considerably high, thus, to avoid high learning losses, we have transformed the $K_d$ values into the log space ($pK_d$) using Eq. (1). The distribution of the $pK_d$ values spans from 5 (10 000 nM) to approximately 11.

$$pK_d = -log_{10}(\frac{K_d}{10^9}) \tag{1}$$

The Davis dataset was split into six different folds using the chemogenomic representative $k$-fold (Monteiro, Oliveira et al., 2022) method, where one of the folds was selected as an independent test set to estimate the performance and generalization capacity of the architecture and the remaining folds to determine the hyperparameters of the binding affinity prediction model. The chemogenomic representative $k$-fold approach takes into consideration the $pK_d$ values distribution, the protein sequences similarity, and the SMILES strings similarity during the splitting process, leading to representative folds in the context of the problem.

### 2.2. 1D binding pocket dataset

The interaction between compounds and proteins results from the recognition and complementarity of particular functional groups (binding sites) in the 3D space. In order to construct the 1D binding pocket dataset, we collected 3D complexes from scPDB (Desaphy et al., 2015), PDBBind (Liu et al., 2017), and BioLiP (Yang et al., 2013a) with binding information available, i.e., complexes with the interacting residues annotated, and associated with drug-like molecules. In order to filter and select biologically relevant ligands, we have used the HET group list from BioLiP (Yang et al., 2013a) and P2Rank (Krivák & Hoksza, 2018), and excluded complexes with less than five binding residues. Single-chain 3D complexes were regarded as single DTI pairs and multiple-chain 3D complexes were split into single-chain interaction pairs. Most of these 3D complex-based DTI pairs, however, correspond to specific fractions of the protein sequence in the 1D space. Thus, to identify the positions of the interacting residues in the whole protein sequence, it was necessary to map these fragments onto the corresponding UniProt (Consortium, 2023) sequence. On that account, we have applied the pairwise sequential local alignment function from Biopython (Cock et al., 2009) to determine the best alignment and identify the corresponding binding positions, where entries with a mismatch greater than 50% between the residues of the original and alignment binding pocket were removed from the dataset. Furthermore, the binding information of single-chain pairs belonging to the same PDB complex and with identical UniProt sequence was unified into single 1D binding pockets.

The binding sites are usually determined based on protein residues whose distance toward ligands is below a certain threshold. However, the definition of a binding pocket is inconsistent across multiple studies or databases, leading to some noise in identifying the correct interacting residues, especially in a 1D representation. Furthermore, residues in the neighborhood of a particular binding residue likely influence its ligandability, which is consistent with the distribution of the binding sites in a 1D representation. These positions are non-consecutive in the 1D sequence, however, they are prone to be concentrated across scattered local binding regions. Hence, in order to reasonably define the 1D binding pocket, the neighborhood of every single interacting position was also taken into account, i.e., for each binding position $p$, the residues within the interval $]p-k, p+k[$, where $k$ was fixed at 3 (Monteiro, Simões et al., 2022), were also considered as binding-related positions. The resulting 1D binding pockets were converted into binary binding vectors of the same length as the corresponding protein sequences, where ones and zeros represent binding and non-binding residues, respectively. Fig. 1 depicts the process to generate the 1D

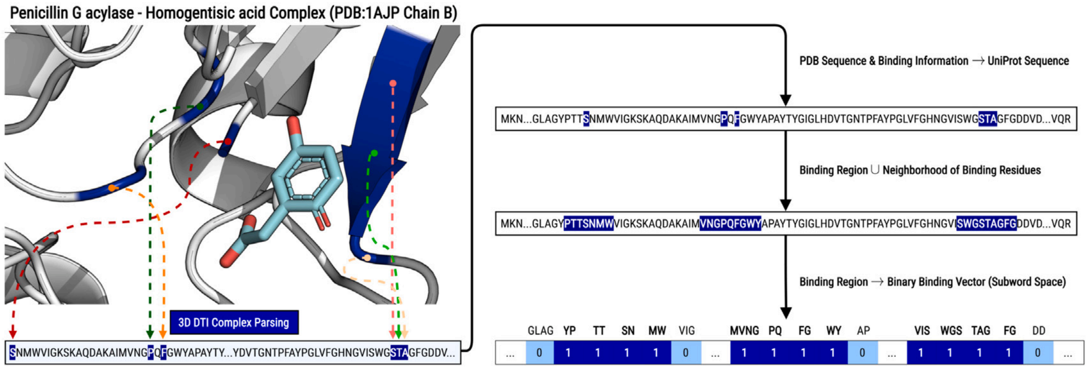



**Fig. 1.** Generation of the 1D binding pocket for the penicillin G acylase - Homogentisic acid complex (PDB: 1AJP chain B). The 3D complex is collected from one of the binding-related databases (scPDB (Desaphy et al., 2015), PDBBind (Liu et al., 2017), or BioLiP (Yang et al., 2013a)) and parsed to the 1D space, in which the protein sequence fragment and the binding positions are retrieved. The 1D binding information is mapped onto the corresponding UniProt (Consortium, 2023) sequence using the biopython (Cock et al., 2009) package, where the neighborhood of each binding position is also taken into consideration. The resulting 1D binding pocket is converted into a binary binding vector, where ones and zeros represent binding and non-binding residues (subwords), respectively.

binding pocket associated with the Penicillin G acylase - Homogentisic acid complex (PDB: 1AJP chain B).

The protein sequences and SMILES strings associated with the resulting 1D binding pocket dataset were processed and encoded based on similar approaches to those applied to the binding affinity dataset. On that account, we have selected only proteins with a length between 30 and 575 subwords and SMILES strings with a sequential length between 10 and 100 chemical tokens.

In order to select the hyperparameters of the 1D binding pocket prediction model, we have split the resulting 1D binding pocket dataset into a 90/10% training/validation dataset ratio. On the other hand, to estimate the binding site prediction model's performance, we have considered the COACH (Yang et al., 2013b) test dataset, which is widely used in several studies related to binding site prediction. The COACH test dataset was processed identically to the 1D binding pocket dataset and duplicated PDB complexes were removed from the 1D binding pocket dataset.

### 2.3. SMILES pre-train MLM dataset

In order to pre-train the SMILES Transformer-Encoder block using the masked language modeling (MLM) approach, we have collected SMILES strings from ChEMBL (Gaulton et al., 2017) associated with small compounds that follow the Lipinski's rule of five (zero violations). Lipinski's rule defines boundaries for certain physicochemical properties, including molecular weight, lipophilicity, polar surface area, number of hydrogen bond acceptors, number of hydrogen bond donors, and number of rotatable bonds, to determine the drug-likeness (orally active) of the molecules.

We have selected only SMILES strings with a sequential length between 10 and 100 chemical tokens and encoded each character into the corresponding integer using the 72-character-integer dictionary. Furthermore, the resulting SMILES pre-train MLM dataset was split into a 90/10% training/validation ratio to select the hyperparameters of the SMILES Transformer-Encoder block.

Table 1 summarizes the statistics of the binding affinity, 1D binding pocket, and SMILES pre-train MLM datasets.

### 2.4. TAG-DTA framework

The TAG-DTA framework simultaneously learns to predict the 1D binding pocket and binding strength of DTIs, where the prediction of the binding sites vector guides and conditions the prediction of DTA. This framework comprises two models, specifically a 1D binding pocket classifier and a binding affinity regressor, and shares three core layers, including lower Transformer-Encoders and a condition-based concatenation block. The architecture uses two parallel Transformer-Encoders to compute contextual embeddings and capture the proteomics and chemical context present in the protein sequences and SMILES strings, respectively, where the SMILES Transformer-Encoder is pre-trained using an MLM approach. The aggregate representation of the SMILES string, which corresponds to the final hidden state of the start token added to the SMILES strings, is concatenated with the resulting protein tokens, followed by conditional and positional encoding. The binding site classifier block, which comprises a Transformer-Encoder with a position-wise FFN, uses the resulting condition-based concatenated tokens as input for binary token labeling learning, predicting the 1D binding pocket. The predicted 1D binding pocket is used to condition the attention mechanism of the Transformer-Encoder of the binding affinity regressor block, which also uses the condition-based concatenated tokens as input, by masking non-binding residues. On that account, it learns the pharmacological space and the interdependencies amongst the binding-related subwords. The resulting aggregate representations of the binding affinity Transformer-Encoder, protein Transformer-Encoder, and SMILES Transformer-Encoder are concatenated and used as input for an FCNN, which outputs the binding affinity measured in terms of $pK_d$. Fig. 2 illustrates the proposed TAG-DTA architecture.

#### 2.4.1. Embedding block

Protein sequences and SMILES strings are tokenized according to the FCS/BPE and character-integer encoding methods, respectively, where each token is converted to a numeric value and each sequence/string is padded up to a maximum value of $N_P/N_S$. Additionally, special start tokens have been added to the beginning of every protein sequence ($T_P$) and SMILES string ($T_S$). In order to map semantic meaning into a geometric space, we have assigned an embedding layer to the protein sequences and SMILES strings, which transforms each token into a learned continuous vector (embedding) with a fixed size of $d^P_{model}$ and $d^S_{model}$ via a learnable dictionary lookup matrix $W^P_{token} \in R^{d^P_{model} \times |V^P|}$ and $W^S_{token} \in R^{d^S_{model} \times |V^S|}$, respectively.

Contrarily to recurrent neural networks, Transformer-Encoders do not have a built-in mechanism to deal with the order of sequences, i.e., they are entirely invariant to sequence order. To provide absolute or relative positional information of the tokens in the sequence to the model, we have included a positional embedding via a learnable dictionary lookup matrix $W^P_{pos} \in R^{d^P_{model} \times N_P}$ and $W^S_{pos} \in R^{d^S_{model} \times N_S}$. The final embeddings $E^P_i$ and $E^S_j$ associated with the $i$th and $j$th input tokens of the protein sequence and SMILES string, respectively, are given by the sum of the token embedding ($E^P_{token_i}$ and $E^S_{token_j}$) and the

**Table 1**
Statistics of collected binding affinity, 1D binding pocket, and SMILES pre-train MLM datasets.

| Dataset | Proteins | Compounds | DTI | pKd = 5 | pKd > 5 | Binding residues | Non-Binding residues |
|---|---|---|---|---|---|---|---|
| ***Binding Affinity*** | | | | | | | |
| Davis Kinase Original | 442 | 72 | 31 824 | 22 400 | 9424 | – | – |
| Davis Kinase Pre-Processed | 423 | 69 | 29 187 | 20 479 | 8708 | – | – |
| ***1D Binding Pocket*** | | | | | | | |
| scPDB ∪ PDBBind ∪ BioLiP | 14 256 | 24 151 | 81 696 | – | – | 1 816 960 (4.4%) | 39 340 137 (95.6%) |
| scPDB ∪ PDBBind ∪ BioLiP[a] | 14 256 | 24 151 | 81 696 | – | – | 2 464 345 (15.9%) | 13 011 113 (84.1%) |
| COACH Test | 402 | 332 | 490 | – | – | 6708 (3.4%) | 190 626 (96.6%) |
| COACH Test[a] | 402 | 332 | 490 | – | – | 11 852 (15.8%) | 63 194 (84.2%) |
| ***SMILES Pre-Train MLM*** | | | | | | | |
| ChEMBL | – | 1 321 328 | – | – | – | – | – |

[a] BPE/FCS Encoding + Neighborhood.

positional embedding ($E^P_{pos_i}$ and $E^S_{pos_j}$), followed by a dropout layer:

$$\begin{aligned}\mathbf{E}^P_i &= \mathbf{dropout}(\mathbf{E}^P_{token_i} + \mathbf{E}^P_{pos_i})\\ \mathbf{E}^S_j &= \mathbf{dropout}(\mathbf{E}^S_{token_j} + \mathbf{E}^S_{pos_j})\end{aligned}, \tag{2}$$

where $E^P_i$, $E^P_{token_i}$, $E^P_{pos_i} \in R^{d^P_{model}}$ and $E^S_j$, $E^S_{token_j}$, $E^S_{pos_j} \in R^{d^S_{model}}$.

The parameters of the token and positional embedding layers associated with the SMILES strings are initialized with the ones learned during the MLM pre-training stage.

### 2.4.2. Transformer-encoder

We use two Transformer-Encoders in parallel to capture the proteomics and chemical context and model the inter-dependency amongst the substructures and molecular components of the protein sequences and SMILES strings, respectively. Transformer-Encoders stack $N$ identical blocks, where each block comprises a multi-head self-attention (MHSA) layer and a position-wise FFN (PWFFN). Residual connections are applied after each subunit followed by layer normalization (LN) to mitigate vanishing gradients. Additionally, dropout is added after each MHSA layer and after each dense layer of the PWFFN to prevent overfitting. Fig. 3 illustrates the architecture of the Transformer-Encoder.

The MHSA layer determines the short and long-term inter-dependencies between the input elements by applying self-attention multiple times in parallel, resulting in a robust and contextual representation for each token of the sequence. This layer takes the input in the form of three parameters, specifically Query ($Q$), Key ($K$), and Value ($V$), and computes attention across $h$ heads of attention. On that account, Q, K, and V are linearly projected and divided into $h$ sub-dimensions, where each $head_{1,\ldots,h}$ computes a weighted sum of $V^{proj}_{1,\ldots,h}$. The attention weights assigned to each element of $V^{proj}_{1,\ldots,h}$ are determined by applying a softmax to the scaled dot-product between $Q^{proj}_{1,\ldots,h}$ and $K^{proj}_{1,\ldots,h}$, where a masking matrix is added before the softmax in order to prevent the model from attending to certain tokens, e.g., $PAD$ tokens. The outputs of each head of attention are concatenated and linearly projected, resulting in the same dimensions as the input $Q$.

$$\begin{aligned}\mathbf{attn}(Q,K,V) &= \mathbf{softmax}(\frac{QK^T}{\sqrt{d_K}} + Mask)V\\ \mathbf{MHSA}(Q,K,V) &= [\mathbf{attn}(Q^{proj}_1, K^{proj}_1, V^{proj}_1); \ldots;\\ &\quad \mathbf{attn}(Q^{proj}_h, K^{proj}_h, V^{proj}_h)]\mathbf{W}^O\\ Q^{proj}_{1,\ldots,h} = QW^Q_{1,\ldots,h},\ & K^{proj}_{1,\ldots,h} = KW^K_{1,\ldots,h},\ V^{proj}_{1,\ldots,h} = VW^V_{1,\ldots,h}\end{aligned}, \tag{3}$$

where $Q \in R^{N_{P/S}\ x\ d^{P/S}_{model}}$, $K \in R^{N_{P/S}\ x\ d^{P/S}_{model}}$, $V \in R^{N_{P/S}\ x\ d^{P/S}_{model}}$, $W^Q_{1,\ldots,h} \in R^{d^{P/S}_{model}\ x\ d_Q}$ are the $Q$ projection matrices, $W^K_{1,\ldots,h} \in R^{d^{P/S}_{model}\ x\ d_K}$ are the $K$ projection matrices, $W^V_{1,\ldots,h} \in R^{d^{P/S}_{model}\ x\ d_V}$ are the $V$ projection matrices, $W^O \in R^{h\ x\ d_V\ x\ d^{P/S}_{model}}$ is the output projection matrix, [; ] denotes concatenation, $d_Q = d_K = d_V = \frac{d^{P/S}_{model}}{h}$, and $Mask \in R^{N_{P/S}\ x\ N_{P/S}}$.

The PWFFN block improves the robustness of the representation of each token and increases the learning capacity of the architecture by projecting the last dimension (position-wise) of the attention outputs. On that account, it is composed of two dense layers, where the first dense layer projects the attention outputs to a higher dimension and the second dense layer projects it back to the initial last dimension.

$$\mathbf{PWFFN}(x) = \mathbf{dropout}(\mathbf{F}_2(\mathbf{dropout}(\mathbf{F}_1(x\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2))), \tag{4}$$

where $W_1 \in R^{d^{P/S}_{model}\ x\ \pi_{P/S}}$, $W_2 \in R^{\pi_{P/S}\ x\ d^{P/S}_{model}}$, $b_1 \in R^{\pi_{P/S}}$, $b_2 \in R^{d^{P/S}_{model}}$, $F$ is the activation function, and $\pi_{P/S}$ is the expansion ratio.

Considering $x^1$ and $x^2$ the outputs of the MHSA layer and the PWFFN block, respectively, the output of the $k$th Transformer-Encoder block can be expressed as:

$$\begin{aligned}x^1_k &= \mathbf{LN}(x^2_{k-1} + \mathbf{dropout}(\mathbf{MHSA}(x^2_{k-1})))\\ x^2_k &= \mathbf{LN}(x^1_k + \mathbf{PWFFN}(x^1_k))\end{aligned}, \tag{5}$$

where $x^1_k$, $x^2_k \in R^{N_{P/S}\ x\ d^{P/S}_{model}}$.

The weights of the SMILES Transformer-Encoder block are initialized with the ones learned during the MLM pre-training step.

### 2.4.3. Condition-based concatenation block

In order to model the interaction and exchange of information between proteins and compounds, it is crucial to represent the pharmacological space of the interaction. On that account, we propose the use of a condition-based concatenation block, which concatenates the $T_S$ token from the SMILES Transformer-Encoder with the resulting protein tokens from the protein Transformer-Encoder. The $T_S$ token previously learns the overall chemical context and inter-dependency amongst the individual units of the SMILES strings and is thus considered an aggregate representation. On the other hand, each resulting token of the protein sequences from the protein Transformer-Encoder is characterized by a robust and contextual representation based on the learned short and long-term dependencies. Considering $X_P$ and $X_S$ the outputs of the protein and SMILES Transformer-Encoder, respectively, the output of the concatenation layer ($X_{DT}$) can be expressed as:

$$X_P = [T_P \parallel C_P],\ X_S = [T_S \parallel C_S] \longrightarrow X_{DT} = [T^{Pool}_S \parallel C_P], \tag{6}$$

where $X_P \in R^{N_P\ x\ d^P_{model}}$, $T_P \in R^{d^P_{model}}$, $C_P \in R^{(N_P-1)\ x\ d^P_{model}}$, $X_S \in R^{N_S\ x\ d^S_{model}}$, $T_S \in R^{d^S_{model}}$, $C_S \in R^{(N_S-1)\ x\ d^S_{model}}$, $T^{Pool}_S \in R^{d^P_{model}}$, and $X_{DT} \in R^{N_P\ x\ d^P_{model}}$.

Considering that the $T_S$ token is used to condition and interact with the protein tokens, i.e., it is not exclusively the attending agent, we have included a conditional embedding via a learnable dictionary lookup matrix $W^{DT}_{cond} \in R^{d^P_{model}\ x\ N_P}$ to distinguish the $T_S$ token from the resulting protein tokens. In order to update the positional information of the tokens in the concatenated DTI representation, we have added a positional embedding via a learnable dictionary lookup matrix $W^{DT}_{pos} \in R^{d^P_{model}\ x\ N_P}$. Following the sum of the conditional embedding and positional embedding, we have included a dropout layer. The final representation $E^{DT}_i$ associated with the $i$th token of the resulting concatenated tokens can be expressed as:

$$\mathbf{E}^{DT}_i = \mathbf{dropout}(\mathbf{E}^{DT}_i + \mathbf{E}^{DT}_{cond_i} + \mathbf{E}^{DT}_{pos_i}), \tag{7}$$

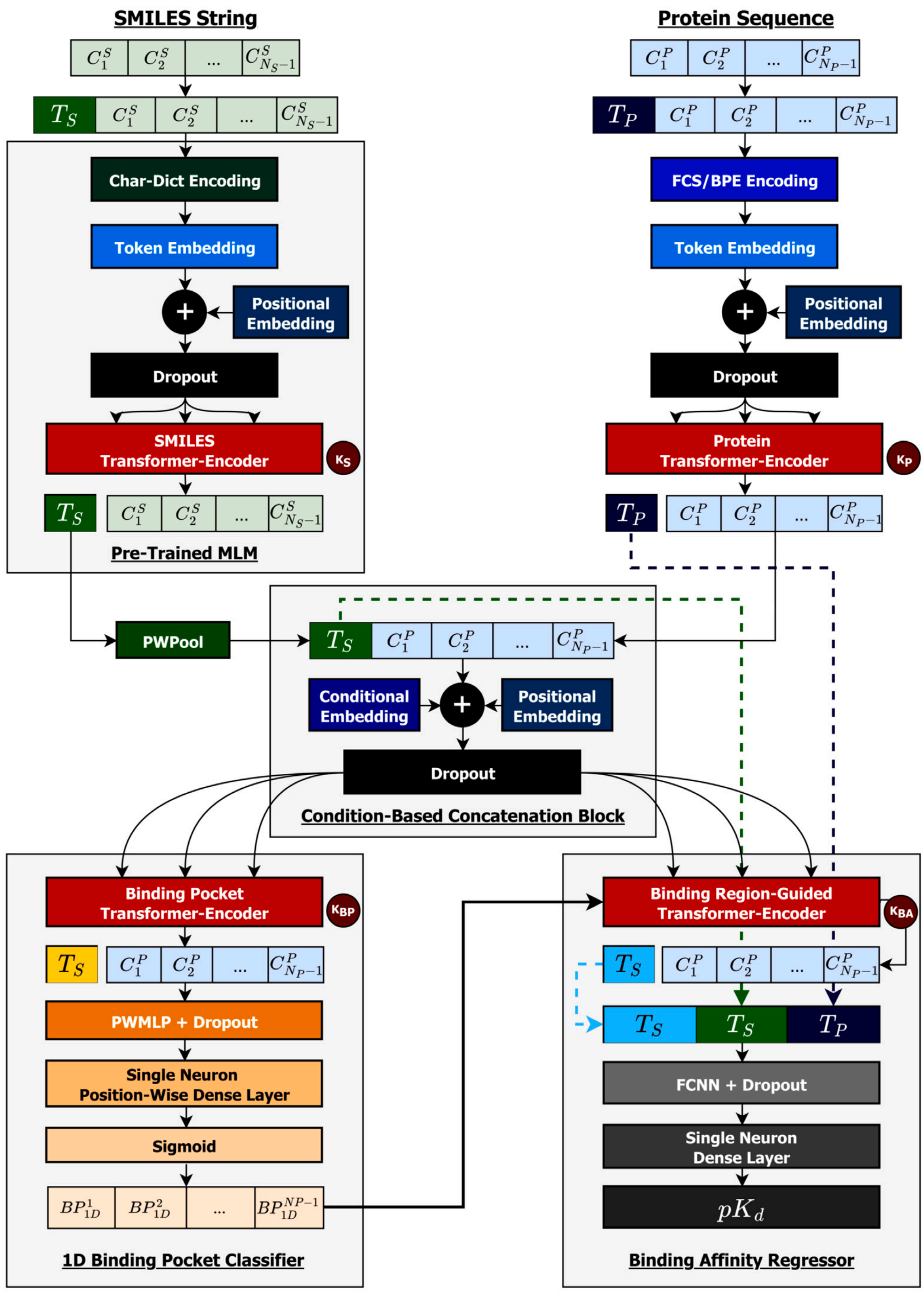



**Fig. 2.** TAG-DTA: End-to-End binding-region-guided transformer-based architecture. Two parallel Transformer-Encoders model the inter-dependencies and capture the proteomics and chemical context present in the protein sequences and SMILES strings, respectively. A condition-based concatenation block concatenates the projected $T_S$ (green) token from the SMILES Transformer-Encoder with the resulting protein tokens from the protein Transformer-Encoder in order to represent the pharmacological space. The resulting concatenated DTI representation is used as input to the 1D binding pocket classifier for binary token labeling, determining the binding nature of each protein subword. The predicted binary binding vector conditions the attention mechanism of the binding-region-guided Transformer-Encoder, resulting in the learning of the interaction context based on binding-related positions. The resulting aggregate representations of the binding affinity Transformer-Encoder (blue $T_S$), protein Transformer-Encoder ($T_P$), and SMILES Transformer-Encoder (green $T_S$) are concatenated and used as input for an FCNN, which outputs the binding affinity measured in terms of $pK_d$.

where $E_i^{DT} \in R^{d^P_{model}}$, $E^{DT}_{cond_i} \in R^{d^P_{model}}$ is the conditional embedding, and $E^{DT}_{pos_i} \in R^{d^P_{model}}$ is the positional embedding.

In order to map the $T_S$ token to the last dimension of the protein tokens ($C_P$), we have added a position-wise pooling (PWPool) block, which comprises a dense layer applied to the last dimension of $T_S$ and an LN layer.

$$\mathbf{PWPool}(T_S) = \mathbf{LN}(\mathbf{F}_{T_S}(\mathbf{W}_{T_S} T_S + \mathbf{b}_{T_S})), \tag{8}$$

where $F$ is the activation function, $W_{T_S} \in R^{d^S_{model} \times d^P_{model}}$, and $b_{T_S} \in R^{d^P_{model}}$

### 2.4.4. 1D binding pocket classifier

The 1D binding pocket classifier learns to identify the binding and non-binding positions within the protein sequence, i.e., it predicts each protein subword as a binding or non-binding spot. This block is

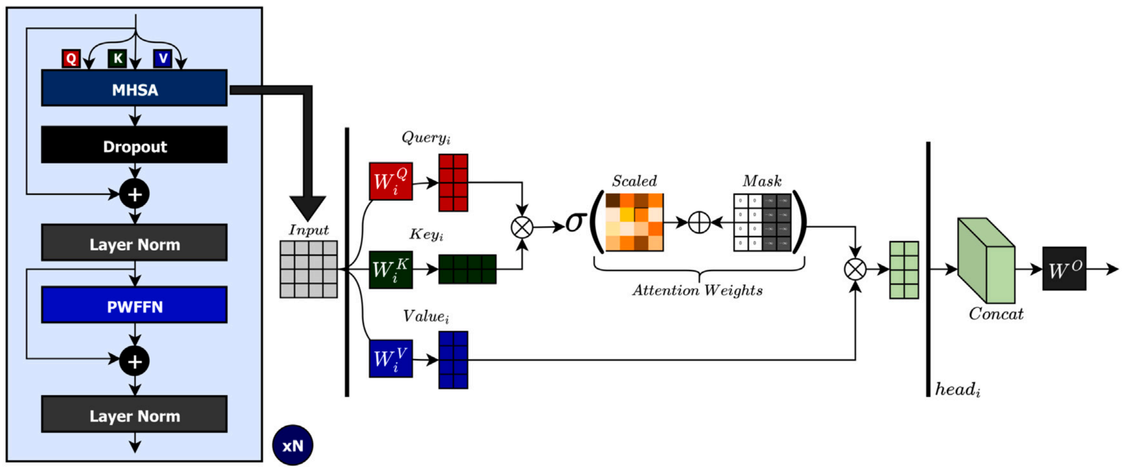



**Fig. 3.** Transformer-Encoder architecture, where each block is composed of an MHSA layer and a PWFFN. The MHSA layer computes self-attention across $h$ heads of attention, where each $head_i$ computes a weighted sum of $V^i_{proj}$. The attention weights are determined by applying a softmax ($\sigma$) to the scaled dot-product between $Q^i_{proj}$ and $K^i_{proj}$, in which $PAD$ tokens are masked. The PWFNN is applied to the last dimension of the MHSA outputs in order to give them an individually more robust representation.

composed of a Transformer-Encoder followed by a position-wise multi-layer perceptron (PWMLP). The binding pocket Transformer-Encoder uses the output of the condition-based concatenation block as input and determines the interaction (and inter-dependencies) between the aggregate representation of the compound and the protein tokens. On that account, the $T_S$ token and the protein tokens attend mutually to each other, where $T_S$ conditions the representation of each token of the protein sequences based on their potential selectivity toward the compound. The resulting protein tokens are fed to the PWMLP, which stacks dense layers applied to the last dimension of the protein tokens (embedding/representation space), in order to increase the learning capacity of this block. Additionally, dropout layers are added after each dense layer of the PWMLP. Following the PWMLP, a single neuron position-wise dense layer is added to classify the binding nature of each protein token, where a sigmoid activation function is applied for binary token labeling. Considering $X_{DT}$ the output of condition-based concatenation block, $X^1_{DT}$ the output of the binding pocket Transformer-Encoder, and $X^2_{DT}$ the output of the PWMLP, the output of the 1D binding pocket classifier ($BP_{1D}$) can be expressed as:

$$\begin{aligned} X^1_{DT} &= \textbf{TransformerEncoder}_{BP}(X_{DT}) \\ X^2_{DT} &= \textbf{PWMLP}(X^1_{DT}) \quad , \\ BP_{1D} &= \sigma(X^2_{DT}\mathbf{W}_{BP_{1D}} + \mathbf{b}_{BP_{1D}}) \end{aligned} \tag{9}$$

where $X_{DT} \in R^{N_P \times d^P_{model}}$, $X^1_{DT} \in R^{N_P \times d^P_{model}}$, $X^2_{DT} \in R^{(N_P-1) \times \pi^2_{DT}}$, $W_{BP_{1D}} \in R^{\pi^2_{DT} \times 1}$, $b_{BP_{1D}} \in R^1$, $BP_{1D} \in \mathbb{Z}_2^{(N_P-1)}$, $\sigma$ is the sigmoid activation function, and $\pi^2_{DT}$ is expansion/contraction ratio of the final dense layer of the PWMLP.

### 2.4.5. Binding affinity regressor

The interaction between active compounds and proteins results from the recognition and complementarity of certain groups (binding regions) and it is supported by the joint action of other individual substructures scattered across the protein and compound. Hence, to effectively predict binding affinity, it is important to consider the proteomics, chemical, and pharmacological spaces. In order to learn the pharmacological context information associated with the interaction space for the prediction of binding affinity, we proposed the use of a binding-region-guided Transformer-Encoder, where the output of the 1D binding pocket classifier guides the attention mechanism and the output of the condition-based concatenation block is used as input. The binding-region-guided Transformer-Encoder architecture is similar to the standard Transformer-Encoder, however, instead of applying global self-attention, it takes into consideration tokens that are potential binding residues using the predicted 1D binding pocket. On that account, the padding masking matrix is combined with the predicted 1D binding pocket, masking non-binding residues and PAD tokens. Thus, this layer learns the inter-dependencies amongst binding-related residues and the interaction between $T_S$ (compound representation) and the binding tokens of the protein sequences. The resulting $T_S$ token is considered an aggregate representation of the pharmacological space and expresses the short and long-term dependencies between the compound and the binding region.

The final hidden states of the aggregate representation of the protein Transformer Encoder, SMILES Transformer-Encoder, and binding-region-guided Transformer-Encoder, respectively, are concatenated, followed by an LN layer, and used as input for an FCNN. Similarly to the PWMLP, we have added dropout after each dense layer of the FCNN. Following the FCNN, a dense layer with a single neuron is applied to predict the binding affinity of the DTI pair measured in terms of the logarithmic-transformed dissociation constant (pK$_d$). Considering $X^1_{Aff}$ the output of the binding-region-guided Transformer-Encoder and $X^2_{Aff}$ the output of the FCNN, the output of the binding affinity regressor ($BF_{pK_D}$) can be expressed as:

$$\begin{aligned} X^1_{Aff} &= \textbf{TransformerEncoder}_{Aff}(X_{DT}) \\ X^2_{Aff} &= \textbf{FCNN}(X^1_{Aff})\ ,\ X^2_{Aff} = [T_{Aff} \parallel C_{Aff}] \\ T_{DTI} &= \textbf{LN}([T_P\ ;\ T_S^{Pool}\ ;\ T_{Aff}]) \quad , \\ BF_{pK_D} &= T_{DTI}\mathbf{W}_{BF_{pK_D}} + \mathbf{b}_{BF_{pK_D}} \end{aligned} \tag{10}$$

where $X_{DT} \in R^{N_P \times d^P_{model}}$, $X^1_{Aff} \in R^{N_P \times d^P_{model}}$, $T_P \in R^{d^P_{model}}$, $T_S^{Pool} \in R^{d^P_{model}}$, $T_{Aff} \in R^{d^P_{model}}$, $C_{Aff} \in R^{(N_P-1) \times d^P_{model}}$, $T_{DTI} \in R^{d^{DTI}_{model}}$ is concatenated representation of the final hidden states of the aggregate representations, $W_{BF_{pK_D}} \in R^{d^{DTI}_{model} \times 1}$, $b_{BF_{pK_D}} \in R^1$, $BF_{pK_D} \in R^1$, $d^{DTI}_{model} = d^P_{model} + d^P_{model} + d^P_{model}$, and [; ] denotes concatenation.

In order to avoid potential negative transfer of information from the 1D binding pocket classifier, i.e., incapable of identifying any binding position for a certain DTI pair, we have added some flexibility to the attention mechanism in the binding-region-guided Transformer-Encoder. On that account, when the predicted binary 1D binding pocket does not contain any value equal to 1 (binding spot), which would lead to the $T_S$ token attending exclusively to itself, we only mask the PAD tokens before the softmax function. Thus, the attention mechanism corresponds to the standard global self-attention, learning the short and long-term inter-dependencies amongst all tokens of the input sequence,

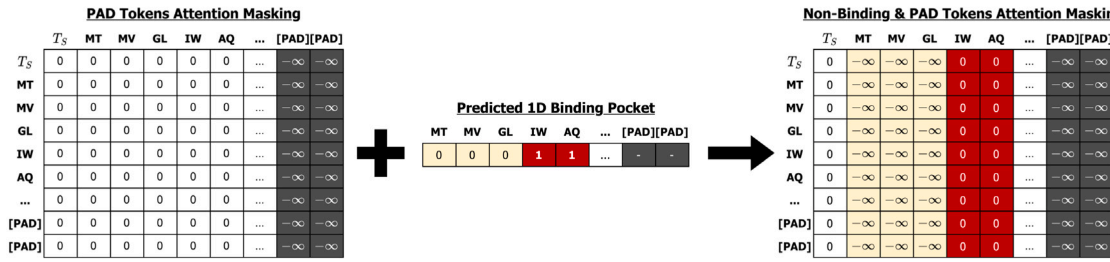



**Fig. 4.** Binding-region-guided attention masking matrix, where the $PAD$ tokens masking matrix is combined with the predicted 1D binding pocket.

except PAD tokens.

$$\textbf{Attention} = \begin{cases} \textbf{Conditioned} \\ \textbf{if} \quad \exists i = 1, \ldots, N_P - N_{PAD} - 1 \; BP_{1D}(i) = 1 \\ \textbf{Global} \\ \textbf{if} \quad \forall i \in \{1, \ldots, N_P - N_{PAD} - 1\}, BP_{1D}(i) = 0 \end{cases}, \tag{11}$$

where $Conditioned$ corresponds to the binding-region-guided attention, $Global$ to the standard global self-attention, and $N_{PAD}$ to the number of padding tokens.

Fig. 4 illustrates the masking matrix applied in the binding-region-guided Transformer-Encoder when the predicted binary 1D binding pocket identifies binding spots (binding-region-guided attention).

#### 2.4.6. SMILES pre-train masked language modeling

MLM leverages a large number of data points in an unsupervised fashion to capture the context and semantics of the input domain. This approach masks certain tokens of the input sequence and designs the model to predict the original tokens based on the unaltered sentence units. On that account, the model needs to learn the statistical and distribution properties as well as the short and long-term dependencies amongst the tokens of the sequence, considering that tokens can have different meanings in different positions. Hence, the model learns deep and multiple representations of the tokens, improving the performance levels in downstream tasks.

In the context of this work, we use the MLM approach to pre-train the SMILES Transformer-Encoder in order to capture the molecular context within the SMILES strings. We follow the traditional MLM masking setup used in BERT (Devlin et al., 2019), which randomly masks 15% of the tokens of the input sequence. The selected tokens are replaced with a special [MASK] token, replaced with a random token from the SMILES dictionary, or remain unaltered based on an 80%, 10%, and 10% probability rate, respectively.

The architecture of the SMILES pre-train MLM includes the SMILES token embedding layer, the SMILES positional embedding layer, the SMILES Transformer-Encoder, a dense layer, and a softmax activation function. The dense layer projects the output of the Transformer-Encoder into the dimension of the SMILES vocabulary $V_S$. The softmax function normalizes the output of the dense layer to a probability distribution over the cardinality of the vocabulary, indicating the likelihood of each token in the vocabulary for each position of the input sequence. Considering $E^S$ the output of the SMILES embedding block and $X_S$ the output of the SMILES Transformer-Encoder, the output of the SMILES pre-train MLM ($X_S^{MLM}$) can be expressed as:

$$\begin{aligned} X_S &= \textbf{TransformerEncoder}_{SMILES}(E^S) \\ X_S^{MLM} &= \textbf{Softmax}(X_S \mathbf{W}_S^{MLM} + \mathbf{b}_S^{MLM}) \end{aligned}, \tag{12}$$

where $E^S \in R^{N_S \times d^S_{model}}$, $X^S \in R^{N_S \times d^S_{model}}$, $W_S^{MLM} \in R^{d^S_{model} \times |V^S|}$, $b_S^{MLM} \in R^{|V^S|}$, and $X_S^{MLM} \in R^{N_S \times |V^S|}$.

Fig. 5 shows the SMILES pre-train MLM architecture.

### 2.5. TAG-DTA training strategy

In order to train the TAG-DTA framework, the training strategy is divided into three stages: (I) pre-training of the 1D binding pocket classifier; (II) training of the binding affinity regressor; and (III) training of the 1D binding pocket classifier. We firstly pre-train the 1D binding pocket subunit, which consists of the three shared blocks, the binding pocket Transformer-Encoder, and the PWMLP, in order to partially converge and optimize the TAG-DTA toward the prediction of the 1D binding pocket, considering that binding sites vector is used to guide and condition the prediction of binding affinity. Following the first stage, we alternatively train the binding affinity subunit, which is composed of the three shared blocks, the binding-region-guided Transformer-Encoder, and the FCNN, and the 1D binding pocket subunit to optimize the TAG-DTA to simultaneously predict the 1D binding pocket and the binding affinity of DTIs. On that account, two different training modes are considered, where only binding pocket or binding affinity-related blocks are trainable during each corresponding prediction task, respectively, i.e., the binding pocket Transformer-Encoder and PWMLP weights are frozen during the training of the binding affinity regressor, and the binding-region-guided Transformer-Encoder and FCNN weights are frozen during the training of the 1D binding pocket classifier. Hence, this training strategy aims to reduce the discrepancy between the 1D binding pocket classification and the binding affinity regression and avoid the negative transfer of information between computationally different tasks yet contextually related. Moreover, instead of considering the convergence of the TAG-DTA architecture at the end of each training epoch, we aim to optimize the framework at the end of each training cycle, which corresponds to the end two of the training modes, i.e., the training of the binding affinity regressor and the training of the 1D binding pocket classifier.

## 3. Experimental setup

### 3.1. SMILES pre-train MLM optimization

The hyperparameters for the SMILES pre-train MLM architecture were determined using the 10% hold-out validation approach, in which the SMILES ChEMBL dataset was randomly split into a 90/10% training/validation ratio. Several parameters were hyperoptimized, including the embedding dimension of the SMILES strings, the number of SMILES Transformer-Encoder layers, the number of attention heads, the PWFNN expansion ratio, the dropout rate, the optimizer learning rate, the optimizer weight drop, the optimizer warm-up ratio, and the batch size. A considerable range of possible values was assigned for each hyperparameter and the search was narrowed down to the best parameter values.

The Gaussian Error Linear Unit (GELU) (Hendrycks & Gimpel, 2020) was selected as the activation function for every layer, except for the final output dense layer which uses a softmax activation. The GELU function is a smoother version of the ReLU activation and can be expressed as:

$$\begin{aligned} &GELU(x) = xP(X \leq x) = x\Phi(x) \\ &\approx \; 0.5x(1 + tanh[\sqrt{2/\pi}(x + 0.044715x^3)]) \end{aligned}, \tag{13}$$

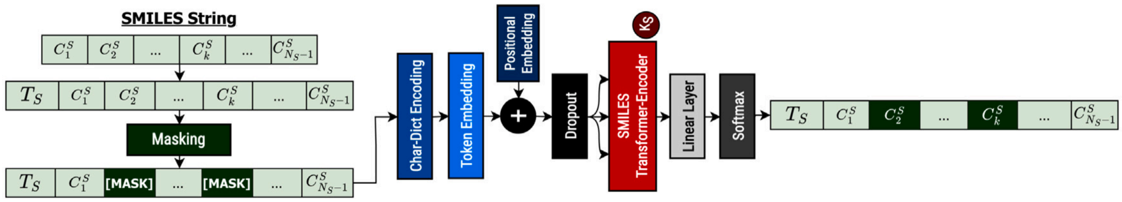



**Fig. 5.** Pre-training of the SMILES Transformer-Encoder using an MLM approach, where the model learns to predict the $[MASK]$ tokens based on the unaltered input units.

where $\Phi(x)$ is the cumulative distribution function for the standard normal distribution (Gaussian) and $P(X) \sim N(0,1)$.

The optimizer function considered to update the network weights in each iteration of the training process was the Rectified Adaptive Moment Estimation (RAdam) (Liu et al., 2021). This algorithm is an enhanced version of the traditional Adam optimizer and it dynamically adjusts the adaptive learning rate based on the underlying divergence of the variance using a variance rectification term. Even though this method usually avoids the need to use a warm-up heuristic, we have included a warm-up learning rate scheduler to promote the training process.

Considering that the SMILES pre-train MLM focuses on the prediction of masked input tokens, the loss function chosen was the categorical cross-entropy (CCE), which measures the divergence between probability distributions for multi-class classification problems. Additionally, all non-masked input chemical tokens were excluded from the calculation of the loss value.

$$CCE = -\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{m} y_{ij} log(p_{ij}), \tag{14}$$

where $n$ is the number of samples, $m$ is the number of classes, $y$ is the true label, and $p(y)$ is the predicted probability.

Moreover, to avoid potential overfitting of the model, we have considered early stopping with a patience of 30 and model checkpoint. The hyperparameter combination that provided the best accuracy of correctly predicting the masked input tokens over the validation set was selected to establish the optimized model. The parameters of the SMILES token embedding layer, SMILES positional embedding layer, and SMILES Transformer-Encoder of the resulting model were used to initialize the parameters of the corresponding blocks in the TAG-DTA framework. Table 2 summarizes the parameter settings for the SMILES pre-train MLM architecture.

**Table 2**
SMILES Pre-Train MLM parameter settings.

| Parameter | Value |
|---|---|
| SMILES strings length ($N_S$) | 101 |
| SMILES Transformer-Encoders | 3 |
| SMILES self-attention heads | 8 |
| SMILES embedding dim | 512 |
| SMILES PWFFN hidden neurons | 2048 |
| Activation function | GELU |
| Activation function (Output) | Softmax |
| Dropout rate | 0.1 |
| Loss function | CCE |
| Optimizer function | RAdam |
| Optimizer learning rate | 1e−03 |
| Optimizer minimum learning rate | 1e−05 |
| Optimizer beta 1 | 0.9 |
| Optimizer beta 2 | 0.999 |
| Optimizer epsilon | 1e−08 |
| Optimizer weight decay | 1e−04 |
| Optimizer warm up proportion | 0.01 |
| Optimizer total steps | 512 500 |
| Batch size | 232 |
| Epochs[a] | 500 |

[a] Initial number of epochs to allow convergence of the model, where early stopping and model checkpoint were applied to avoid overfitting.

### 3.2. TAG-DTA optimization

The hyperoptimization approach applied to the TAG-DTA architecture is based on 10% hold-out validation and chemogenomic representative $k$-fold cross-validation, where the former is applied to the binding sites dataset and the latter to the binding affinity dataset. These two methods are combined with grid-search, where early stopping and model checkpoint are considered over the training cycles in order to avoid overfitting. We have hyperoptimized several parameters: embedding dimension of the protein sequences, number of protein Transformer-Encoder layers, number of protein Transformer-Encoder attention heads, protein Transformer-Encoder PWFFN expansion ratio, number of binding-pocket Transformer-Encoder layers, number of binding-pocket Transformer-Encoder attention heads, binding-pocket Transformer-Encoder PWFFN expansion ratio, PWMLP number of layers, PWMLP hidden neurons, number of binding-region-guided Transformer-Encoder layers, number of binding-region-guided Transformer-Encoder attention heads, binding-region-guided Transformer-Encoder PWFFN expansion ratio, FCNN number of layers, FCNN hidden neurons, 1D binding pocket optimizer learning rate, 1D binding pocket optimizer weight decay, binding affinity optimizer learning rate, binding Affinity optimizer weight decay, SMILES Transformer-Encoder optimizer learning rate, SMILES Transformer-Encoder optimizer weight decay, dropout rate, number of epochs for the pre-training of the binding sites classifier, number of epochs for the training of the binding affinity regressor, and number of epochs for the training of the binding sites classifier. We initially considered a wide range of values for each hyperparameter and then narrowed the search range around the best parameter values.

The GELU (Hendrycks & Gimpel, 2020) activation function was selected for every layer, except for the final output dense layer of the 1D binding pocket classifier and binding affinity regressor, which uses a sigmoid function and a linear activation, respectively. The optimizer selected to update the weights of the pre-trained layers, 1D binding pocket classifier, and binding affinity regressor was the RAdam (Liu et al., 2021). Additionally, we have considered a step decay learning rate scheduler in the case of the optimizers used in the 1D binding pocket and binding affinity training modes. The step decay scheduler drops the learning rate by a factor of 0.5 every 25 training cycles.

$$Step\_Decay(TC) = LR_{init} * factor^{floor(TC/TC_{drop})}, \tag{15}$$

where $LR_{init}$ is the initial learning rate, $factor$ is the learning rate dropping factor, $TC$ is the current training cycle, and $TC_{drop}$ is the number of training cycles that the learning rate keeps constant.

Considering that the context of the problem focuses on a classification and a regression task, the loss function selected for the prediction of the 1D binding pocket and the binding affinity of DTIs was the binary cross entropy (BCE) and the mean squared error (MSE), respectively. BCE measures the divergence between two probability distributions and is a special case of the CCE. Additionally, considering the imbalanced nature of the distribution of binding and non-binding positions, we have introduced different class weights.

$$BCE = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(p_i) + (1-y_i)\log(1-p_i)\right], \tag{16}$$

**Table 3**
TAG-DTA parameter settings.

| Parameter | Value |
|---|---|
| SMILES strings length ($N_S$) | 101 |
| SMILES Transformer-Encoders | 3 |
| SMILES self-attention heads | 8 |
| SMILES embedding dim | 512 |
| SMILES PWFFN hidden neurons | 2048 |
| Protein sequences length ($N_P$) | 576 |
| Protein Transformer-Encoders | 3 |
| Protein self-attention heads | 4 |
| Protein embedding dim | 256 |
| Protein PWFFN hidden neurons | 1024 |
| Binding pocket Transformer-Encoders | 1 |
| Binding pocket self-attention heads | 4 |
| Binding pocket PWFFN hidden neurons | 1024 |
| PWMLP dense layers | 3 |
| PWMLP hidden neurons | [128,64,32] |
| Binding-region-guided Transformer-Encoders | 1 |
| Binding-region-guided self-attention heads | 4 |
| Binding-region-guided PWFFN hidden neurons | 1024 |
| FCNN dense layers | 3 |
| FCNN hidden neurons | [1536,1536,1536] |
| Activation function | GELU |
| Activation function (1D Binding Pocket Output) | Sigmoid |
| Activation function (Binding Affinity Output) | Linear |
| Dropout rate | 0.1 |
| 1D binding pocket loss function | BCE |
| 1D binding pocket loss class weights | 0: 0.4, 1: 0.6 |
| Binding affinity loss function | MSE |
| SMILES pre-trained layers optimizer | RAdam, LR:1e−05, Beta 1: 0.9, Beta 2: 0.999, Epsilon: 1e−08, Weight Decay: 1e−05 |
| 1D binding pocket classifier optimizer | RAdam, LR:1e−04, Beta 1: 0.9, Beta 2: 0.999, Epsilon: 1e−08, Weight Decay: 1e−05 |
| Binding affinity regressor optimizer | RAdam, LR:1e−04, Beta 1: 0.9, Beta 2: 0.999, Epsilon: 1e−08, Weight Decay: 1e−05 |
| Batch size | 32 |
| 1D binding pocket pre-train epochs | 20 |
| Binding affinity train epochs | 3 |
| 1D binding pocket train epochs | 1 |
| Epochs[a] | 500 |

[a] Initial number of epochs to allow convergence of the model, where early stopping and model checkpoint over the training cycles were applied to avoid overfitting.

where $n$ is the number of samples, $y$ is the true label, and $p_i(y)$ is the predicted probability. On the other hand, MSE measures the average squared difference between the predicted values and the real values.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2, \tag{17}$$

where $n$ is the number of samples, $y$ is the true value, and $\hat{y}$ is the predicted value.

The hyperparameter combination that provided the best average MSE score over the validation sets in the case of binding affinity and the best Matthew's correlation coefficient (MCC), which computes the correlation between two binary variables, in the case of the 1D binding pocket was selected to establish the optimized model. Table 3 summarizes the parameter settings for the TAG-DTA framework.

$$MCC = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}, \tag{18}$$

where TP are the True positives, TN are the True negatives, FP are the False positives, and FN are the False negatives.

To validate and assess the prediction efficiency of binding affinity of the proposed TAG-DTA architecture, we have evaluated and compared the performance with various state-of-the-art binding affinity regression and binary DTI classification benchmarks: KronRLS (Pahikkala et al., 2014), SimBoost (He et al., 2017), Sim-CNN-DTA (Shim et al., 2021), DeepDTA (Öztürk et al., 2018), DeepCDA (Abbasi et al., 2020), TransformerCPI (Chen et al., 2020), HyperAttentionDTI (Zhao et al., 2022), DTITR (Monteiro, Oliveira et al., 2022), and all the different formulations of the GraphDTA (Nguyen et al., 2020). In the case of the binary DTI classification benchmarks, such as TransformerCPI (Chen et al., 2020) and HyperAttentionDTI (Zhao et al., 2022), we have transformed them into regression models by modifying their final layers, i.e., the output of the last layer was replaced with a single neuron dense layer, and by altering their loss function to MSE. Furthermore, we have applied the FCS/BPE encoding approach to the protein sequences in the research works where the proteins are represented by their 1D amino acid sequence in order to ensure fairness in the comparisons. To further validate the binding affinity performance of the proposed framework and increase the fairness in the comparisons with the state-of-the-art benchmarks, we have also compared the results using the standard experimental settings of these benchmarks, i.e., the same split method of the Davis binding affinity dataset.

Considering the existing unexplored space in the drug discovery domain and the importance of the models to generalize toward unknown subsets of the proteomics and/or chemical representation spaces, we have evaluated and compared the performance of the proposed TAG-DTA framework in three experimental settings, specifically novel target proteins, novel compounds, and novel compound-target pairs. On that account, we have randomly split the Davis binding affinity dataset into three groups of five folds each based on these three restrictions,

**Table 4**
Binding affinity prediction results over the Davis independent test set.

| Method | Protein Rep. | Compound Rep. | ↓ MSE | ↓ RMSE | ↑ CI | ↑ $r^2$ | ↑ Spearman |
|---|---|---|---|---|---|---|---|
| ***Baseline methods*** | | | | | | | |
| KronRLS (Pahikkala et al., 2014) | Smith-Waterman | PubChem-Sim | 0.443 | 0.665 | 0.847 | 0.473 | 0.624 |
| GraphDTA-GCNNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.311 | 0.558 | 0.883 | 0.630 | 0.681 |
| TransformerCPI (Chen et al., 2020) | 1D-Subseq | 1D | 0.291 | 0.539 | 0.852 | 0.511 | 0.507 |
| GraphDTA-GATNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.286 | 0.535 | 0.881 | 0.660 | 0.688 |
| SimBoost (He et al., 2017) | Smith-Waterman | PubChem-Sim | 0.277 | 0.526 | 0.891 | 0.670 | 0.694 |
| GraphDTA-GAT-GCN (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.269 | 0.518 | 0.874 | 0.680 | 0.670 |
| Sim-CNN-DTA (Shim et al., 2021) | Smith-Waterman | PubChem-Sim | 0.266 | 0.516 | 0.884 | 0.683 | 0.674 |
| GraphDTA-GINConvNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.238 | 0.488 | 0.899 | 0.717 | **0.741** |
| HyperAttentionDTI (Zhao et al., 2022) | 1D-Subseq | 1D | 0.227 | 0.477 | 0.890 | 0.729 | 0.690 |
| DeepDTA (Öztürk et al., 2018) | 1D-Subseq | 1D | 0.215 | 0.464 | 0.891 | 0.743 | 0.691 |
| DeepCDA (Abbasi et al., 2020) | 1D-Subseq | 1D | 0.208 | 0.457 | 0.895 | 0.752 | 0.689 |
| DTITR (Monteiro, Oliveira et al., 2022) | 1D-Subseq | 1D | 0.192 | 0.438 | 0.907 | 0.771 | 0.712 |
| ***Proposed method*** | | | | | | | |
| **TAG-DTA** | 1D-Subseq | 1D | **0.185** | **0.430** | **0.917** | **0.780** | 0.729 |

i.e., proteins, compounds, and DTIs absent from the corresponding training set, respectively.

Apart from evaluating the performance and the generalization capacity of the proposed TAG-DTA framework in the prediction of the binding affinity of DTI pairs, it is critical to explore the efficacy and contribution of certain blocks in the learning capacity of the TAG-DTA architecture, especially considering the dual nature of the prediction process associated with this model. On that account, we have explored various alternatives of the TAG-DTA model, specifically the contribution to the learning capacity of the architecture by pre-training the SMILES Transformer-Encoder blocks using the MLM approach, the performance of the TAG-DTA architecture to exclusively predict the 1D binding pocket, and the contribution of the prediction of the 1D binding pocket for the prediction of binding affinity, i.e., the performance of the TAG-DTA architecture to exclusively predict binding affinity without limiting the attention mechanism of the Transformer-Encoder associated with the binding affinity regression block.

We used Python 3.9.6 and Tensorflow 2.8.0 to develop all models and the experiments were run on AMD Ryzen 9 3900X and GeForce RTX 3080 10 GB.

## 4. Results and discussion

### *4.1. Binding affinity prediction performance evaluation*

In the context of drug discovery and drug repositioning, it is essential to properly assess the target selectivity of potential leads. Thus, establishing models capable of accurately predicting unbiased bioactivities associated with the interaction of biologically relevant targets and active small molecules is pivotal on the road to new insights in the DTI field. In order to validate the performance of the proposed TAG-DTA architecture in the prediction of binding affinity, we have evaluated and compared the prediction efficiency with various state-of-the-art binding affinity regression and binary classification models. Table 4 reports the binding affinity prediction results over the Davis independent test set in terms of MSE, RMSE, CI, $r^2$, and Spearman rank correlation (See Section 1.1 of Supplementary material for more details regarding the evaluation metrics).

The results demonstrate that the proposed TAG-DTA framework achieved the highest performance in terms of MSE (0.185), RMSE (0.430), CI (0.917), and $r^2$ (0.780) compared to the state-of-the-art benchmarks. Thus, it exceeds the other models in its ability to correctly predict the binding affinity values (lower MSE and RMSE scores) and distinguish the binding strength rank order across DTI pairs (higher CI score). Furthermore, the significant increase in the CI metric shows the superior capacity of the architecture to correctly assess the target selectivity, which is crucial in identifying potential leads and differentiating primary from secondary interactions. Moreover, these findings are consistent with the results of Table 5, which reports the binding affinity prediction results over the Davis dataset using the original split methodology of the state-of-the-art research works. The TAG-DTA showed a significant increase in performance across all metrics, specifically MSE (0.199 ± 0.003), RMSE (0.446 ± 0.003), CI (0.898 ± 0.001), $r^2$ (0.752 ± 0.004), and Spearman rank correlation (0.710 ± 0.002), and lower standard deviation values compared to the state-of-the-art benchmarks, which furthers validates the efficiency of the proposed architecture in the prediction of binding affinity and shows superior learning stability. Additionally, in order to assess whether there are statistical differences between the performance of the proposed architecture and the baseline models, we carried out pairwise comparisons utilizing the paired t-test, with a 95% confidence interval, applied to the CI scores. The proposed architecture, TAG-DTA, outperformed the baselines with statistical significance on the CI score, resulting in $P$-value $< 0.05$ & $< 0.01$ across all pairwise comparisons. In particular, the TAG-DTA versus DTITR pairwise comparison yielded a $P$-value of 8.047e−04.

In order to properly predict DTA it is essential to learn the inter-dependency of the sequential and structural units of each binding component and the inter-associations that revolve around the binding-related substructures (pharmacological space). The majority of the baseline methods, however, either focus on learning individual representations of the proteins and compounds, which are usually combined for the inferring process, or only take into consideration the mutual interaction space for the prediction of binding affinity. Moreover, the mere use of Transformers, such as TransformerCPI (Chen et al., 2020), or attention mechanisms, e.g., HyperAttentionDTI (Zhao et al., 2022), do not necessarily guarantee a performance increase, considering that the short and long-distance interactions within the proteins and compounds are crucial for the DTA prediction performance. On that account, DTITR (Monteiro, Oliveira et al., 2022) showed superior performance to all previous state-of-the-art benchmarks since it models the intra-associations within each individual binding component and the inter-dependency between the involving interacting components. However, DTITR (Monteiro, Oliveira et al., 2022) does not model the inter-dependency amongst binding-related tokens or the interaction between the compound and the binding region. TAG-DTA overcomes this limitation by employing a binding-region-guided Transformer-Encoder to learn the pharmacological space based on the short and long-term dependencies between the compound and the binding region and the inter-dependency within the binding region. Additionally, it takes into consideration the magnitude of the local regions of each binding component and their intra-associations. Overall, the results demonstrate that the TAG-DTA model is properly learning the proteomics, chemical, and pharmacological context of the proteins, compounds, and DTIs, respectively, and that guiding the learning of the pharmacological space based on potential binding positions leads to improved DTA prediction performance. Fig. 6 depicts the predictions from the TAG-DTA model against the actual binding affinity values for the Davis

**Table 5**
Binding affinity prediction results over the Davis dataset using the original split methodology, where the standard deviations are given in parentheses.

| Method | Protein Rep. | Compound Rep. | ↓ MSE | ↓ RMSE | ↑ CI | ↑ $r^2$ | ↑ Spearman |
|---|---|---|---|---|---|---|---|
| ***Baseline methods*** | | | | | | | |
| KronRLS (Pahikkala et al., 2014)[a] | Smith-Waterman | PubChem-Sim | 0.379 | 0.616 | 0.871 (0.001) | – | – |
| GraphDTA-GCNNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.322 (0.047) | 0.566 (0.039) | 0.850 (0.020) | 0.598 (0.059) | 0.636 (0.032) |
| TransformerCPI (Chen et al., 2020) | 1D-Subseq | 1D | 0.285 (0.024) | 0.533 (0.022) | 0.839 (0.017) | 0.493 (0.043) | 0.472 (0.023) |
| GraphDTA-GAT-GCN (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.285 (0.004) | 0.534 (0.004) | 0.862 (0.006) | 0.644 (0.005) | 0.656 (0.009) |
| GraphDTA-GATNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.284 (0.012) | 0.533 (0.012) | 0.872 (0.002) | 0.646 (0.015) | 0.681 (0.010) |
| SimBoost (He et al., 2017)[a] | Smith-Waterman | PubChem-Sim | 0.282 | 0.531 | 0.872 (0.002) | – | – |
| Sim-CNN-DTA (Shim et al., 2021) | Smith-Waterman | PubChem-Sim | 0.276 (0.008) | 0.525 (0.008) | 0.869 (0.006) | 0.656 (0.010) | 0.666 (0.011) |
| GraphDTA-GINConvNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.256 (0.004) | 0.506 (0.004) | 0.875 (0.005) | 0.680 (0.005) | 0.708 (0.011) |
| HyperAttentionDTI (Zhao et al., 2022) | 1D-Subseq | 1D | 0.241 (0.005) | 0.491 (0.005) | 0.879 (0.002) | 0.699 (0.006) | 0.680 (0.004) |
| DeepDTA (Öztürk et al., 2018) | 1D-Subseq | 1D | 0.235 (0.006) | 0.485 (0.006) | 0.871 (0.006) | 0.707 (0.007) | 0.670 (0.014) |
| DeepCDA (Abbasi et al., 2020) | 1D-Subseq | 1D | 0.232 (0.004) | 0.482 (0.004) | 0.879 (0.003) | 0.710 (0.005) | 0.680 (0.005) |
| DTITR (Monteiro, Oliveira et al., 2022) | 1D-Subseq | 1D | 0.216 (0.006) | 0.465 (0.006) | 0.880 (0.005) | 0.730 (0.007) | 0.681 (0.008) |
| ***Proposed method*** | | | | | | | |
| **TAG-DTA** | 1D-Subseq | 1D | **0.199 (0.003)** | **0.446 (0.003)** | **0.898 (0.001)** | **0.752 (0.004)** | **0.710 (0.002)** |

[a] Benchmarks results from Öztürk et al. (2018).

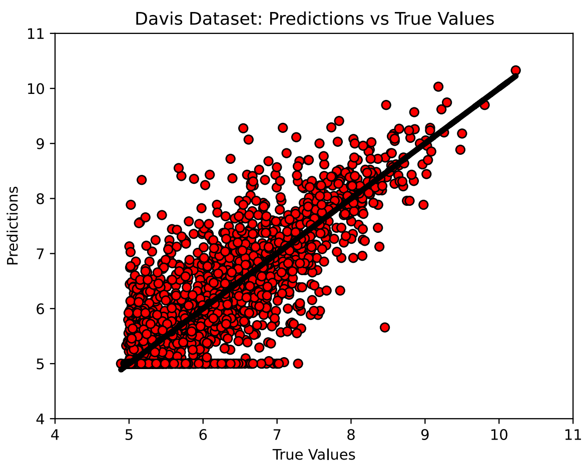



**Fig. 6.** TAG-DTA binding affinity predictions against the true values for the Davis affinity testing set, where the black diagonal line corresponds to the reference line (*predicted* = *true value*).

independent test set, where is it possible to observe a significant density around the *predicted* = *true value* reference line (perfect model).

Considering the importance of the models to generalize toward unknown subsets of the proteomics and/or chemical representation spaces, we have explored and evaluated the performance of the proposed framework in the prediction of binding affinity for three experimental settings, specifically novel compounds, novel proteins, and novel protein-compounds pairs. Table 6 reports the binding affinity prediction results over a 5-fold random split of the Davis affinity dataset for these three experimental settings in terms of MSE, RMSE, and CI. Regarding the novel compound and novel protein-compound pairs settings, TAG-DTA achieved lower values of MSE (0.600 ± 0.064 and 0.609 ± 0.057) and RMSE (0.775 ± 0.042 and 0.780 ± 0.036), and higher CI scores (0.715 ± 0.031 and 0.670 ± 0.058) compared to the state-of-the-art benchmarks. However, we find no significant difference between the proposed architecture and attention-based benchmarks, specifically DTITR (Monteiro, Oliveira et al., 2022), HyperAttentionDTI (Zhao et al., 2022), and TransformerCPI (Chen et al., 2020), in these two experimental settings. In the case of the novel protein experimental setting, the proposed TAG-DTA framework achieved an MSE of 0.372 ± 0.044, an RMSE of 0.549 ± 0.035, and a CI score of 0.833 ± 0.013, demonstrating superior and competitive performance to generalize in unknown subsets of the proteomics space. Additionally, all models showed improved performance in the novel protein setting compared to the novel compound and novel protein-compound pair settings, which is consistent with the Kinase representability properties of the Davis affinity dataset. Furthermore, the proposed TAG-DTA model showed overall lower standard deviations amongst the validation sets across all three experimental settings, which demonstrates improved learning stability. In spite of these findings, it is noteworthy that no statistical significance was observed with respect to the CI score in any pairwise comparison ($P$-value > 0.05), except in the case of the GraphDTA (Nguyen et al., 2020) architecture ($P$-value < 0.05 for all formulations of the architecture), across all three experimental settings.

### *4.2. TAG-DTA ablation study*

In order to further validate the TAG-DTA architecture, we have explored various alternatives for the TAG-DTA model, specifically (i) TAG-DTA architecture without pre-training the SMILES Transformer-Encoder, (ii) TAG-DTA architecture without the binding affinity regression block, and (iii) TAG-DTA architecture without the 1D binding pocket classification block. Table 7 reports the binding affinity and 1D binding pocket prediction results over the Davis and COACH test set, respectively, for the various alternatives of the TAG-DTA model.

To properly assess the efficacy of pre-training the SMILES Transformer-Encoder using an MLM approach, which leverages a vast amount of data points in an unsupervised fashion to learn the context and semantics of the chemical domain, we have evaluated the model prediction efficiency with and without initializing the SMILES Transformer-Encoder related layers weights with the ones learned during the MLM pre-training step (See Section 2.1 of the Supplementary material for more details regarding the masked token prediction results over the training and validation sets). The TAG-DTA architecture with the pre-trained SMILES Transformer-Encoder resulted in overall better performance in terms of the MSE (0.185), RMSE (0.430), CI (0.917), $r^2$ (0.780), and Spearman (0.729) when compared to the TAG-DTA architecture without pre-training the SMILES Transformer-Encoder (MSE - 0.190, RMSE - 0.436, CI - 0.912, $r^2$ - 0.773, and Spearman - 0.721) for the prediction of binding affinity. Moreover, in the case of the 1D binding pocket prediction, the TAG-DTA with the pre-trained SMILES Transformer-Encoder demonstrated a significant increase in performance in terms of balanced accuracy (87.10%), recall (81.32%), precision (68.17%), F1-score (74.16%), and MCC (0.692) when compared to the TAG-DTA architecture without pre-training the SMILES Transformer-Encoder (balanced accuracy - 85.99%, recall - 79.25%, precision - 67.80%, F1-score - 72.93%, and MCC - 0.676). These results show that pre-training the SMILES Transformer-Encoder using an MLM approach to learn the context and semantics of the chemical domains improves the overall learning capacity of the TAG-DTA architecture and increases the discriminative power and robustness of the aggregated representation of the SMILES strings, which is not only used to interact with the protein tokens (and condition their

**Table 6**
Binding affinity prediction results over a 5-fold random split of the Davis affinity dataset for three experimental settings: novel compounds, novel proteins, and novel protein-compound pairs.

| Method | Protein Rep. | Compound Rep. | ↓ MSE | ↓ RMSE | ↑ CI |
|---|---|---|---|---|---|
| Novel compound | | | | | |
| ***Baseline methods*** | | | | | |
| GraphDTA-GCNNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.808 (0.134) | 0.896 (0.074) | 0.628 (0.056) |
| GraphDTA-GATNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.793 (0.101) | 0.889 (0.056) | 0.594 (0.068) |
| GraphDTA-GAT-GCN (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.763 (0.116) | 0.871 (0.066) | 0.628 (0.093) |
| GraphDTA-GINConvNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.636 (0.052) | 0.797 (0.032) | 0.685 (0.056) |
| DeepDTA (Öztürk et al., 2018) | 1D-Subseq | 1D | 0.641 (0.054) | 0.800 (0.034) | 0.696 (0.058) |
| DeepCDA (Abbasi et al., 2020) | 1D-Subseq | 1D | 0.620 (0.057) | 0.786 (0.036) | 0.708 (0.048) |
| HyperAttentionDTI (Zhao et al., 2022) | 1D-Subseq | 1D | 0.612 (0.075) | 0.782 (0.049) | 0.706 (0.053) |
| TransformerCPI (Chen et al., 2020) | 1D-Subseq | 1D | 0.606 (0.072) | 0.778 (0.046) | 0.687 (0.092) |
| DTITR (Monteiro, Oliveira et al., 2022) | 1D-Subseq | 1D | 0.604 (0.052) | 0.777 (0.032) | 0.710 (0.027) |
| ***Proposed method*** | | | | | |
| **TAG-DTA** | 1D-Subseq | 1D | **0.600 (0.064)** | **0.775 (0.042)** | **0.715 (0.031)** |
| Novel protein | | | | | |
| ***Baseline methods*** | | | | | |
| GraphDTA-GCNNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.588 (0.048) | 0.766 (0.031) | 0.744 (0.010) |
| GraphDTA-GATNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.652 (0.021) | 0.807 (0.013) | 0.733 (0.014) |
| GraphDTA-GAT-GCN (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.579 (0.064) | 0.759 (0.042) | 0.739 (0.018) |
| GraphDTA-GINConvNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.495 (0.031) | 0.703 (0.022) | 0.770 (0.011) |
| DeepDTA (Öztürk et al., 2018) | 1D-Subseq | 1D | 0.399 (0.062) | 0.630 (0.049) | 0.807 (0.026) |
| DeepCDA (Abbasi et al., 2020) | 1D-Subseq | 1D | 0.396 (0.045) | 0.628 (0.037) | 0.815 (0.015) |
| HyperAttentionDTI (Zhao et al., 2022) | 1D-Subseq | 1D | 0.396 (0.057) | 0.628 (0.045) | 0.814 (0.020) |
| TransformerCPI (Chen et al., 2020) | 1D-Subseq | 1D | 0.385 (0.046) | 0.620 (0.037) | 0.785 (0.033) |
| DTITR (Monteiro, Oliveira et al., 2022) | 1D-Subseq | 1D | 0.380 (0.052) | 0.615 (0.043) | 0.827 (0.017) |
| ***Proposed method*** | | | | | |
| **TAG-DTA** | 1D-Subseq | 1D | **0.372 (0.044)** | **0.549 (0.035)** | **0.833 (0.013)** |
| Novel protein-compound pair | | | | | |
| ***Baseline methods*** | | | | | |
| GraphDTA-GCNNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.815 (0.129) | 0.900 (0.073) | 0.575 (0.066) |
| GraphDTA-GATNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.788 (0.101) | 0.886 (0.057) | 0.573 (0.104) |
| GraphDTA-GAT-GCN (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.791 (0.107) | 0.887 (0.062) | 0.598 (0.072) |
| GraphDTA-GINConvNet (Nguyen et al., 2020) | 1D-Subseq | Graph | 0.698 (0.087) | 0.834 (0.053) | 0.674 (0.079) |
| DeepDTA (Öztürk et al., 2018) | 1D-Subseq | 1D | 0.630 (0.058) | 0.793 (0.036) | 0.664 (0.053) |
| DeepCDA (Abbasi et al., 2020) | 1D-Subseq | 1D | 0.627 (0.053) | 0.791 (0.033) | 0.665 (0.053) |
| HyperAttentionDTI (Zhao et al., 2022) | 1D-Subseq | 1D | 0.620 (0.070) | 0.787 (0.046) | 0.667 (0.055) |
| TransformerCPI (Chen et al., 2020) | 1D-Subseq | 1D | 0.615 (0.073) | 0.784 (0.050) | 0.652 (0.082) |
| DTITR (Monteiro, Oliveira et al., 2022) | 1D-Subseq | 1D | 0.612 (0.085) | 0.782 (0.051) | 0.665 (0.062) |
| ***Proposed method*** | | | | | |
| **TAG-DTA** | 1D-Subseq | 1D | **0.609 (0.057)** | **0.780 (0.036)** | **0.670 (0.058)** |

**Table 7**
Binding affinity and 1D binding pocket prediction results over the Davis and COACH test set, respectively, for the various alternatives of the TAG-DTA model: (I) TAG-DTA without pre-training the SMILES Transformer-Encoder related block; (II) TAG-DTA without the binding affinity regression block; (III) TAG-DTA without the 1D binding pocket classification block.

| 1D binding pocket prediction | | | | | | | |
|---|---|---|---|---|---|---|---|
| Method | Protein Rep. | Compound Rep. | ↑ Balanced accuracy | ↑ Recall | ↑ Precision | ↑ F1-Score | ↑ MCC |
| TAG-DTA - I | 1D-Subseq | 1D | 85.99 | 79.25 | 67.80 | 72.93 | 0.676 |
| TAG-DTA - II | 1D-Subseq | 1D | **87.99** | **81.45** | **73.70** | **77.38** | **0.730** |
| TAG-DTA | 1D-Subseq | 1D | 87.18 | 81.26 | 68.82 | 74.52 | 0.696 |
| Binding affinity prediction | | | | | | | |
| Method | Protein Rep. | Compound Rep. | ↓ MSE | ↓ RMSE | ↑ CI | ↑ $r^2$ | ↑ Spearman |
| TAG-DTA - I | 1D-Subseq | 1D | 0.190 | 0.436 | 0.912 | 0.773 | 0.721 |
| TAG-DTA - III | 1D-Subseq | 1D | 0.199 | 0.446 | 0.906 | 0.763 | 0.713 |
| TAG-DTA | 1D-Subseq | 1D | **0.185** | **0.430** | **0.917** | **0.780** | **0.729** |

representation) but also for the prediction of binding affinity. Moreover, to evaluate whether there are statistical differences between the performance of TAG-DTA and TAG-DTA architecture without pre-training the SMILES Transformer-Encoder, we executed an unpaired t-test, with a 95% confidence interval. This analysis was applied to both the CI and MCC scores, and it encompassed data obtained from five random runs. The findings indicate that the proposed TAG-DTA model outperformed the TAG-DTA architecture without pre-training the SMILES Transformer-Encoder, demonstrating statistical significance with a $P$-value of 8.677e−05 (<0.05 & <0.01) for the CI score and a $P$-value of 5.498e−05 (<0.05 & <0.01) for the MCC score.

Regarding the prediction efficiency of the TAG-DTA model without the binding affinity regression block, i.e., to exclusively predict the 1D binding pocket, the performance obtained over the COACH test dataset is slightly superior in terms of balanced accuracy (87.99%), recall (81.45%), precision (73.70%), F1-score (77.38%), and MCC (0.730) when compared to the TAG-DTA model with the dual nature in the inferring process (balanced accuracy - 87.18%, recall - 81.26%, precision - 68.82%, F1-score - 74.52%, and MCC - 0.696). These results suggest that using TAG-DTA to simultaneously predict the 1D binding pocket and binding affinity of DTIs reduces the learning capacity of the TAG-DTA to predict the 1D binding pocket, which is expected considering that predicting the 1D binding pocket is computationally complex due to the imbalanced nature of binding and non-binding positions and that these two prediction tasks are computationally different. In this context, the performance disparities, particularly in terms of the MCC score, exhibit statistical significance, yielding a $p$-value of 6.086e−07, which falls below the significance thresholds of 0.05 and 0.01. Nevertheless, the small performance gap demonstrates that the proposed TAG-DTA architecture is capable of converging toward two different prediction tasks, and the overall good performance in the prediction of the 1D binding pocket indicates the viability of the TAG-DTA to learn and identify binding-related positions.

Additionally, we have evaluated the contribution of the prediction of the 1D binding pocket for the prediction of binding affinity by training the TAG-DTA without the 1D binding pocket classification block. The prediction efficiency of the TAG-DTA architecture without the 1D binding pocket classification block resulted in significantly worse performance in terms of the MSE (0.199), RMSE (0.446), CI (0.906), $r^2$ (0.763), and Spearman (0.713) when compared to the TAG-DTA architecture with the 1D binding pocket classification block (MSE - 0.185, RMSE - 0.430, CI - 0.917, $r^2$ - 0.780, and Spearman - 0.729). These results demonstrate that using the predicted 1D binding pocket to guide and condition the attention mechanism of the Transformer-Encoder of the binding affinity regression block improves the discriminative power of the final aggregated representation hidden states associated with the interaction space and the capacity of this block to learn the pharmacological context information associated with the interaction space. Moreover, it indicates that learning the pharmacological space based on the short and long-term dependencies between the compound and the binding region (and the intra-associations within the binding region) leads to improved performance, which is in agreement with the fact that protein binding pockets are crucial in the interaction mechanism involved in DTIs. Furthermore, the superior binding affinity prediction performance of the proposed TAG-DTA architecture demonstrates that the 1D binding pocket and binding affinity prediction models are contextually related and that the model is capable of converging toward two different prediction tasks. In addition to these findings, the TAG-DTA architecture outperformed TAG-DTA without the 1D binding pocket classification block with statistical significance, resulting in a $P$-value of 5.856e−07 (<0.05 & <0.01) for the CI score.

Overall, the use of an end-to-end binding-region-guided Transformer-based architecture, which simultaneously predicts the 1D binding pocket and binding strength of DTIs, demonstrates that actively integrating information about binding sites in the training process is crucial to properly learning the pharmacological space of the interaction and leads to improved binding affinity prediction performance. Additionally, it shows the capacity of the attention mechanisms of the Transformer-Encoders to learn the inter-dependencies between the compound and the protein tokens for the prediction of the 1D binding pocket, and the ability of the Transformer-Encoders to learn robust and discriminative aggregated representations of the proteins, compounds, and pharmacological space for the prediction of binding affinity.

### 4.3. DTI and model understanding

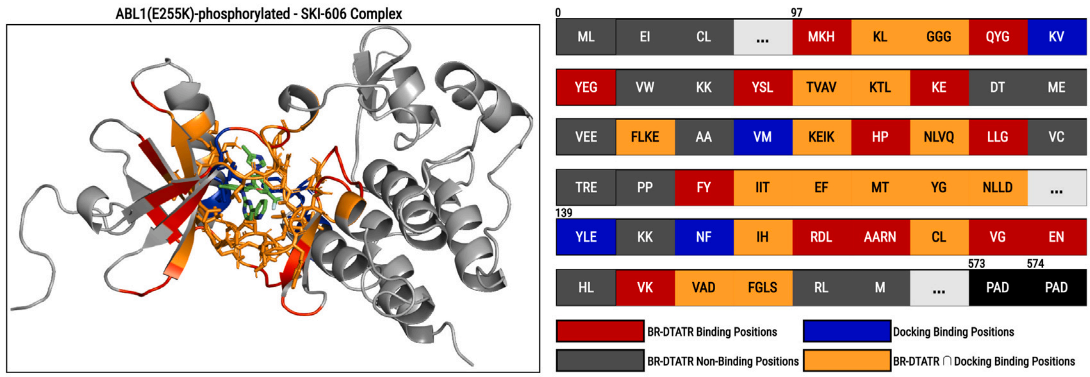
In spite of the increasing performance of the models to correctly predict the binding strength or binary association of DTIs, it is crucial to understand the model prediction and the overall importance of the input intra-associations and inter-dependencies to the model in the context of drug discovery. Moreover, given that DTIs revolve around specific substructures between the binding components and are supported by individual substructures within each interacting element, providing potential model understanding may lead to significant findings in the DTI domain. Considering the nature of the attention blocks, which give information about the overall importance of the input components (and their associations) to the model, the TAG-DTA architecture provides four different levels of attention: (I) protein sequences self-attention; (II) SMILES strings self-attention; (III) 1D binding pocket condition-based self-attention; and (IV) binding-region-guided conditioned-based self-attention. The first and second levels of attention provide information about the overall importance of the individual units (substructures) and intra-associations of the protein sequences (proteomics context) and SMILES strings (chemical context), respectively. The third level of attention gives information about the inter-associations between the compound representation and protein tokens (and their intra-associations) for the prediction of the 1D binding pocket, i.e., how the chemical information affects the overall importance of the individual units and intra-associations of protein sequences for the prediction of the binding pocket. The fourth level of attention provides information about the inter-associations between the compound representation and binding-related protein tokens, and the inter-associations between the binding tokens, i.e., it reflects the interaction and selectivity to the ligand based on the binding pocket.

The visualization and analysis of the different levels of attention provide an increased model and DTI understanding, however, they do not show explicit evidence of potential key residues within the protein sequences for the binding process, which is essential to promote the identification and selection of potential leads and understanding of the biological functions of proteins and mechanisms involved in DTIs. On that account, the proposed TAG-DTA may provide increasing DTI and model understanding considering that it predicts the 1D binding pocket of the DTIs, which also conditions the prediction of the binding affinity. In order to further validate the reliability of the TAG-DTA in the prediction of the 1D binding pocket and that it is capable of providing potential evidence for understanding the model prediction, we have selected the ABL1(E255K)-phosphorylated - SKI-606 DTI pair, which does not have the 3D interaction space available or annotated. We selected potential binding positions (≤5 Å) based on the research work of Monteiro, Simões et al. (2022), where the 3D interaction space was explored and thoroughly assessed using guided docking, and compared with the predicted 1D binding pocket from the TAG-DTA architecture. Fig. 7 depicts the 3D receptor–ligand complex and the TAG-DTA 1D binding pocket for the ABL1(E255K)-phosphorylated - SKI-606 DTI pair.

These visual results show that the TAG-DTA 1D binding pocket identifies almost all the docking binding hits (≤5 Å), which further validates the capacity of the TAG-DTA architecture to learn and identify binding-related positions. Moreover, the other TAG-DTA binding hits are in the neighborhood of the docking binding pocket, where the spacial position of some of these hits (Fig. 7(a)) suggests that they bear relation to conserved regions of the proteins or other potential interaction pockets/subpockets, e.g., some of these hits are near $\beta$-strands, which are usually important for the structure and function of the protein. Overall, these findings demonstrate that TAG-DTA is capable of providing increasing DTI and model understanding, and improves the validity of the learning stage of the pharmacological space based on binding-related positions.

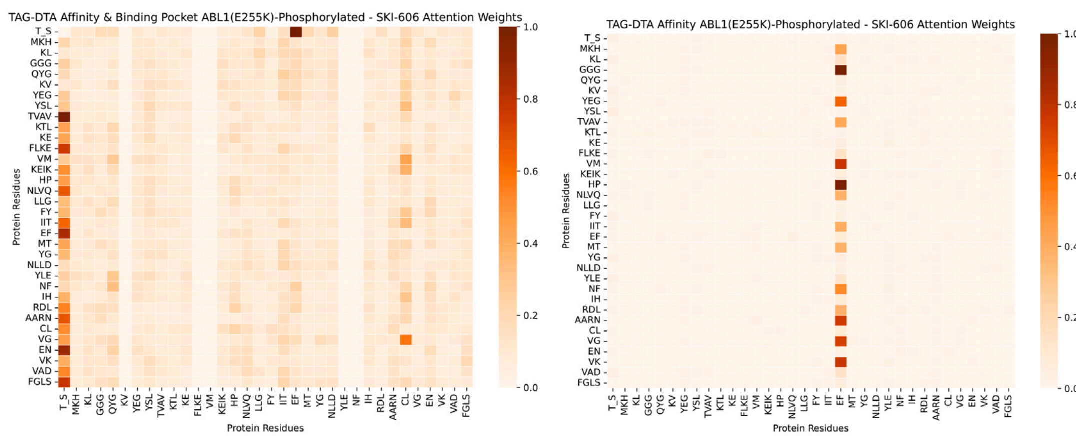
In order to further validate the contribution of the 1D binding pocket block in the learning process of TAG-DTA, we have generated heat maps for the fourth level of attention, specifically for the attention related to the inter-associations between the compound representation and binding-related protein tokens (and inter-associations between binding subwords). We have selected the ABL1(E255K)-phosphorylated - SKI-606 DTI pair for examination, and we have conducted a visual analysis of the attention disparities across binding hits between TAG-DTA without the 1D binding pocket classification block and TAG-DTA with the dual nature incorporated into the inference process. Fig. 8 illustrates the attention heat maps for ABL1(E255K)-phosphorylated - SKI-606 DTI pair across the two TAG-DTA configurations, where only subwords predicted as binding hits or related to docking binding hits were considered for visualization.

These visual observations indicate that TAG-DTA without the 1D binding pocket classification block (Fig. 8(b)) is not attending, i.e., assigning significance, to the majority of protein subwords associated




(a) (b)

**Fig. 7.** SKI-606 in complex with ABL1(E255K)-phosphorylated. (a) Annotated 3D complex obtained from docking (Monteiro, Simões et al., 2022). (b) TAG-DTA 1D binding pocket. The docking binding sites (≤5), TAG-DTA binding positions, TAG-DTA non-binding positions, and matched binding positions are represented by the blue, red, gray, and orange colors, respectively.




(a) (b)

**Fig. 8.** Attention maps associated with the binding-region-guided Transformer-Encoder for the SKI-606 in complex with ABL1(E255K)-phosphorylated. (a) TAG-DTA. (b) TAG-DTA without the 1D binding pocket classification block. The attention weights were normalized across all the positions for each head of attention and the maximum value was selected for visualization.

with binding. Furthermore, it fails to learn the inter-associations between the compound representation ($T_S$) and binding-related protein tokens. Notably, the $T_S$ token does not direct its attention to any binding-related tokens, except for the EF motif. Consequently, this architecture configuration introduces bias into the predictions, resulting in the estimation of potential DTIs based on redundant sites. Conversely, TAG-DTA (Fig. 8(a)), which incorporates a dual nature in the inference process, assigns significance to all binding hits (except for certain docking-related tokens) and comprehensively learns the pharmacological space of the interaction based on the inter-associations amongst binding-related substructures. Furthermore, TAG-DTA demonstrates a lower absolute prediction error for DTA prediction in comparison to TAG-DTA without the 1D binding pocket classification block. In the particular case of ABL1(E255K)-phosphorylated - SKI-606, the $pK_d$ values are 10.33 for the experiment, 10.23 for TAG-DTA, and 10.04 for TAG-DTA without the 1D binding pocket classification block. These findings underscore the critical importance of actively integrating information about binding sites throughout the training process to effectively learn the pharmacological space of the interaction and to enhance the predictive performance of DTA. Moreover, they highlight the capability of TAG-DTA to provide reasonable evidence for understanding model predictions.

## 5. Conclusion

In this research study, we propose a binding-region-guided Transformed-based architecture (TAG-DTA) to simultaneously predict the 1D binding pocket and the binding affinity in terms of the logarithmic-transformed dissociation constant ($pK_d$) of DTI pairs, where the prediction of the 1D binding vector conditions the prediction of DTA. The architecture comprises two models, specifically a 1D binding pocket classifier and a binding affinity regressor, and shares three core layers, including lower Transformer-Encoders and a condition-based concatenation block. The prediction of the 1D binding pocket conditions the attention mechanism of the binding affinity Transformer-Encoder, resulting in the exchange of information between the proteomics and chemical domains over binding-related residues. To perform our experiments, we collected DTIs with binding information annotated from various binding-related databases and DTIs with binding affinity available from the Davis kinase binding affinity dataset. We compare the performance of the proposed TAG-DTA model with different state-of-the-art benchmarks and in different experimental setups based on unknown subsets of the proteomics and chemical representation spaces.

The proposed model yielded better results than state-of-the-art benchmarks for the prediction of binding affinity, resulting in lower

MSE and RMSE values and higher CI and $r^2$ scores in the Davis independent test dataset, and lower MSE and RMSE values and higher CI, $r^2$, and Spearman scores in the original Davis split folds. These results demonstrate the model's ability to accurately predict the value of binding strength and to properly assess the target selectivity (distinguish the rank order of binding strength between the DTI pairs). In the novel compounds, novel proteins, and novel protein-compounds pairs experimental settings, TAG-DTA achieved lower values of MSE and RMSE, and higher CI scores when compared to the benchmarks, demonstrating a superior capacity to generalize toward unknown subsets of the proteomics and chemical representation spaces. Furthermore, the TAG-DTA architecture is shown to efficiently learn the proteomics and chemical context of the proteins and compounds, respectively, and that learning the pharmacological space based on the short and long-term dependencies between the compound and the binding region (and the intra-associations within the binding region) leads to improved DTA prediction performance.

We have explored the influence of different blocks on the prediction efficiency of the TAG-DTA architecture. It was found that pre-training the SMILES Transformer-Encoder using an MLM approach resulted in overall better performance in the prediction of the binding affinity and 1D binding pocket as it increases the discriminating power and robustness of the aggregated representation of the SMILES string, i.e., it improves the learning capacity of this block to capture chemical context. In addition, it was further demonstrated that conditioning the attention mechanism of the binding affinity Transformer-Encoder based on the predicted 1D binding pocket leads to significantly improved DTA prediction performance. Moreover, the results validated that combining computationally different yet contextually related tasks (1D binding pocket classification and binding affinity regression) is crucial for the DTI domain representation and DTA prediction performance.

TAG-DTA provides different levels of potential DTI and prediction understanding due to the nature of the attention layers, which give information about the overall importance of the input components (and their associations) to the model. Moreover, it showed increasing model understanding due to the dual nature of the prediction process, specifically due to the prediction of the 1D binding pocket, which conditions the prediction of the binding affinity and presents explicit evidence of potential key regions within the protein sequences, including in DTI pairs without the 3D complex annotated.

The major contribution of this study is an efficient and novel end-to-end binding-region-guided Transformer-based architecture capable of simultaneously predicting the 1D binding pocket and binding affinity of DTI pairs, where binding information is actively integrated into the training process. Moreover, it models the inter-dependency of the proteomics, chemical, and binding-region-related pharmacological spaces, and provides increasing DTI and model understanding due to the nature of the attention blocks and prediction of the 1D binding pocket.

Considering the improved performance due to the pre-training of the SMILES Transformer-Encoder using an MLM approach, future developments of this research will focus on extending the pre-training step to the protein sequences. However, it is computationally complex and requires considerable resources. Moreover, the state-of-the-art in binding pocket prediction focuses on using 3D information, thus, exploring different levels of structural information or representations might lead to interesting findings. Furthermore, the number of DTIs with binding affinity measured in $K_d$ is reduced compared to other bioactivities, hence, investigating bioactivity-related domain adaptation methods can potentially result in increased prediction performance, considering that deep learning-based architectures perform significantly better when the dataset becomes larger.

## CRediT authorship contribution statement

**Nelson R.C. Monteiro:** Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft. **José L. Oliveira:** Conceptualization, Writing – review & editing, Project administration. **Joel P. Arrais:** Conceptualization, Writing – review & editing, Project administration.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Funding

This work is funded by the FCT - Foundation for Science and Technology, I.P./MCTES through national funds (PIDDAC), within the scope of CISUC R&D Unit - UIDB/00326/2020 or project code UIDP/00326/2020. Nelson R. C. Monteiro gratefully acknowledges FCT for supporting this research work under the Ph.D. grant 2020.04741.BD.

## Data availability

The data and code used in this research are freely available at the github repository mentioned in the main article.

## Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.eswa.2023.122334.

## References

Abbasi, K., Razzaghi, P., Poso, A., Amanlou, M., Ghasemi, J. B., & Masoudi-Nejad, A. (2020). DeepCDA: deep cross-domain compound-protein affinity prediction through LSTM and convolutional neural networks. *Bioinformatics*, *36*(17), 4633–4642. http://dx.doi.org/10.1093/bioinformatics/btaa544.

Agamah, F. E., Mazandu, G. K., Hassan, R., Bope, C. D., Thomford, N. E., Ghansah, A., & Chimusa, E. R. (2019). Computational/in silico methods in drug target and lead prediction. *Briefings in Bioinformatics*, *21*(5), 1663–1675. http://dx.doi.org/10.1093/bib/bbz103.

Altschul, S. F., Madden, T. L., Schäffer, A. A., Zhang, J., Zhang, Z., Miller, W., & Lipman, D. J. (1997). Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Research*, *25*(17), 3389–3402. http://dx.doi.org/10.1093/nar/25.17.3389.

Bongers, B. J., IJzerman, A. P., & Van Westen, G. J. (2019). Proteochemometrics - recent developments in bioactivity and selectivity modeling. *Drug Discovery Today: Technologies*, *32–33*, 89–98. http://dx.doi.org/10.1016/j.ddtec.2020.08.003, Artificial Intelligence.

Brylinski, M., & Skolnick, J. (2007). A threading-based method (FINDSITE) for ligand-binding site prediction and functional annotation. *Proceedings of the National Academy of Sciences of the United States of America*, *105*(1), 129–134. http://dx.doi.org/10.1073/pnas.0707684105, PMC2224172.

Capra, J. A., Laskowski, R. A., Thornton, J. M., Singh, M., & Funkhouser, T. A. (2009). Predicting protein ligand binding sites by combining evolutionary sequence conservation and 3D structure. *PLoS Computational Biology*, *5*(12), Article e1000585. http://dx.doi.org/10.1371/journal.pcbi.1000585.

Castelvecchi, D. (2016). Can we open the black box of AI? *Nature*, *538*, 20–23. http://dx.doi.org/10.1038/538020a.

Chen, L., Tan, X., Wang, D., Zhong, F., Liu, X., Yang, T., Luo, X., Chen, K., Jiang, H., & Zheng, M. (2020). TransformerCPI: improving compound–protein interaction prediction by sequence-based deep learning with self-attention mechanism and label reversal experiments. *Bioinformatics*, *36*(16), 4406–4414. http://dx.doi.org/10.1093/bioinformatics/btaa524.

Cock, P. J. A., Antao, T., Chang, J. T., Chapman, B. A., Cox, C. J., Dalke, A., Friedberg, I., Hamelryck, T., Kauff, F., Wilczynski, B., & de Hoon, M. J. L. (2009). Biopython: freely available python tools for computational molecular biology and bioinformatics. *Bioinformatics*, *25*(11), 1422–1423. http://dx.doi.org/10.1093/bioinformatics/btp163.

Coelho, E. D., Arrais, J. P., & Oliveira, J. L. (2016). Computational discovery of putative leads for drug repositioning through drug-target interaction prediction. *PLoS Computational Biology*, *12*(11), Article e1005219. http://dx.doi.org/10.1371/journal.pcbi.1005219.

Consortium, T. U. (2023). UniProt: the universal protein knowledgebase in 2023. *Nucleic Acids Research*, *51*(D1), D523–D531. http://dx.doi.org/10.1093/nar/gkac1052.

Cui, Y., Dong, Q., Hong, D., & Wang, X. (2019). Predicting protein-ligand binding residues with deep convolutional neural networks. *BMC Bioinformatics*, *20*(1), 93. http://dx.doi.org/10.1186/s12859-019-2672-1.

Davis, M. I., Hunt, J. P., Herrgard, S., Ciceri, P., Wodicka, L. M., Pallares, G., Hocker, M., Treiber, D. K., & Zarrinkar, P. P. (2011). Comprehensive analysis of kinase inhibitor selectivity. *Nature biotechnology*, *29*(11), 1046–1051. http://dx.doi.org/10.1038/nbt.1990.

Desaphy, J., Bret, G., Rognan, D., & Kellenberger, E. (2015). Sc-PDB: a 3D-database of ligandable binding sites—10 years on. *Nucleic Acids Research*, *43*(D1), D399–D404. http://dx.doi.org/10.1093/nar/gku928.

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. arXiv:1810.04805v2.

D'Souza, S., Prema, K. V., & Balaji, S. (2020). Machine learning models for drug–target interactions: current knowledge and future directions. *Drug Discovery Today*, *25*(4), 748–756. http://dx.doi.org/10.1016/j.drudis.2020.03.003.

Gaulton, A., Hersey, A., Nowotka, M., Bento, A. P., Chambers, J., Mendez, D., Mutowo, P., Atkinson, F., Bellis, L. J., Cibrián-Uhalte, E., Davies, M., Dedman, N., Karlsson, A., Magariños, M. P., Overington, J. P., Papadatos, G., Smit, I., & Leach, A. R. (2017). The ChEMBL database in 2017. *Nucleic Acids Research*, *45*(D1), D945–D954. http://dx.doi.org/10.1093/nar/gkw1074.

Gilson, M. K., Liu, T., Baitaluk, M., Nicola, G., Hwang, L., & Chong, J. (2016). BindingDB in 2015: A public database for medicinal chemistry, computational chemistry and systems pharmacology. *Nucleic Acids Research*, *44*(D1), D1045–D1053. http://dx.doi.org/10.1093/nar/gkv1072.

Hanson, J., Paliwal, K. K., Litfin, T., Yang, Y., & Zhou, Y. (2020). Getting to know your neighbor: Protein structure prediction comes of age with contextual machine learning. *Journal of Computational Biology*, *27*(5), 796–814. http://dx.doi.org/10.1089/cmb.2019.0193.

He, T., Heidemeyer, M., Ban, F., Cherkasov, A., & Ester, M. (2017). SimBoost: a read-across approach for predicting drug–target binding affinities using gradient boosting machines. *Journal of Cheminformatics*, *9*(1), 24. http://dx.doi.org/10.1186/s13321-017-0209-z.

Hendrycks, D., & Gimpel, K. (2020). Gaussian error linear units (GELUs). arXiv:1606.08415v4.

Huang, K., Xiao, C., Glass, L. M., & Sun, J. (2021). MolTrans: Molecular interaction transformer for drug–target interaction prediction. *Bioinformatics*, *37*(6), 830–836. http://dx.doi.org/10.1093/bioinformatics/btaa880.

Jiménez, J., Škalič, M., Martínez-Rosell, G., & De Fabritiis, G. (2018). KDEEP: Protein-ligand absolute binding affinity prediction via 3D-convolutional neural networks. *Journal of Chemical Information and Modeling*, *58*(2), 287–296. http://dx.doi.org/10.1021/acs.jcim.7b00650.

Jones, D., Kim, H., Zhang, X., Zemla, A., Stevenson, G., Bennett, W. F. D., Kirshner, D., Wong, S. E., Lightstone, F. C., & Allen, J. E. (2021). Improved protein–ligand binding affinity prediction with structure-based deep fusion inference. *Journal of Chemical Information and Modeling*, *61*(4), 1583–1592. http://dx.doi.org/10.1021/acs.jcim.0c01306.

Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., Li, Q., Shoemaker, B. A., Thiessen, P. A., Yu, B., Zaslavsky, L., Zhang, J., & Bolton, E. E. (2023). PubChem 2023 update. *Nucleic Acids Research*, *51*(D1), D1373–D1380. http://dx.doi.org/10.1093/nar/gkac956.

Krivák, R., & Hoksza, D. (2018). P2rank: machine learning based tool for rapid and accurate prediction of ligand binding sites from protein structure. *Journal of Cheminformatics*, *10*(1), 39. http://dx.doi.org/10.1186/s13321-018-0285-8.

Kumar, S., & Kim, M.-h. (2021). SMPLIP-score: predicting ligand binding affinity from simple and interpretable on-the-fly interaction fingerprint pattern descriptors. *Journal of Cheminformatics*, *13*(1), 28. http://dx.doi.org/10.1186/s13321-021-00507-1.

Landrum, G. (2021). RDKit: Open-source cheminformatics. URL http://www.rdkit.org.

Lee, I., Keum, J., & Nam, H. (2019). DeepConv-DTI: Prediction of drug-target interactions via deep learning with convolution on protein sequences. *PLoS Computational Biology*, *15*(6), Article e1007129. http://dx.doi.org/10.1371/journal.pcbi.1007129.

Liu, L., Jiang, H., He, P., Chen, W., Liu, X., Gao, J., & Han, J. (2021). On the variance of the adaptive learning rate and beyond. arXiv:1908.03265v4.

Liu, Z., Su, M., Han, L., Liu, J., Yang, Q., Li, Y., & Wang, R. (2017). Forging the basis for developing protein–ligand interaction scoring functions. *Accounts of Chemical Research*, *50*(2), 302–309. http://dx.doi.org/10.1021/acs.accounts.6b00491.

London, A. J. (2019). Artificial intelligence and black-box medical decisions: Accuracy versus explainability. *Hastings Center Report*, *49*(1), 15–21. http://dx.doi.org/10.1002/hast.973.

Luo, M., Wang, X. S., Roth, B. L., Golbraikh, A., & Tropsha, A. (2014). Application of quantitative structure–activity relationship models of 5-HT1A receptor binding to virtual screening identifies novel and potent 5-HT1A ligands. *Journal of Chemical Information and Modeling*, *54*(2), 634–647. http://dx.doi.org/10.1021/ci400460q.

Ma, J., Sheridan, R. P., Liaw, A., Dahl, G. E., & Svetnik, V. (2015). Deep neural nets as a method for quantitative structure-activity relationships. *Journal of Chemical Information and Modeling*, *55*(2), 263–274. http://dx.doi.org/10.1021/ci500747n.

Meli, R., Anighoro, A., Bodkin, M. J., Morris, G. M., & Biggin, P. C. (2021). Learning protein-ligand binding affinity with atomic environment vectors. *Journal of Cheminformatics*, *13*(1), 59. http://dx.doi.org/10.1186/s13321-021-00536-w.

Monteiro, N. R., Oliveira, J. L., & Arrais, J. P. (2022). DTITR: End-to-end drug–target binding affinity prediction with transformers. *Computers in Biology and Medicine*, *147*, Article 105772.

Monteiro, N. R. C., Ribeiro, B., & Arrais, J. P. (2021). Drug-target interaction prediction: End-to-end deep learning approach. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, *18*(6), 2364–2374. http://dx.doi.org/10.1109/TCBB.2020.2977335.

Monteiro, N. R. C., Simões, C. J. V., Ávila, H. V., Abbasi, M., Oliveira, J. L., & Arrais, J. P. (2022). Explainable deep drug–target representations for binding affinity prediction. *BMC Bioinformatics*, *23*(1), 237. http://dx.doi.org/10.1186/s12859-022-04767-y.

Morgan, H. L. (1965). The generation of a unique machine description for chemical structures-A technique developed at chemical abstracts service. *Journal of Chemical Documentation*, *5*(2), 107–113. http://dx.doi.org/10.1021/c160017a018.

Nguyen, T., Le, H., Quinn, T. P., Nguyen, T., Le, T. D., & Venkatesh, S. (2020). GraphDTA: Predicting drug-target binding affinity with graph neural networks. *Bioinformatics*, http://dx.doi.org/10.1093/bioinformatics/btaa921.

Öztürk, H., Özgür, A., & Ozkirimli, E. (2018). DeepDTA: deep drug-target binding affinity prediction. *Bioinformatics*, *34*(17), i821–i829. http://dx.doi.org/10.1093/bioinformatics/bty593.

Pahikkala, T., Airola, A., Pietilä, S., Shakyawar, S., Szwajda, A., Tang, J., & Aittokallio, T. (2014). Toward more realistic drug-target interaction predictions. *Briefings in Bioinformatics*, *16*(2), 325–337. http://dx.doi.org/10.1093/bib/bbu010.

Peng, L., Zhu, W., Liao, B., Duan, Y., Chen, M., Chen, Y., & Yang, J. (2017). Screening drug-target interactions with positive-unlabeled learning. *Scientific Reports*, *7*(1), 8087. http://dx.doi.org/10.1038/s41598-017-08079-7.

Rifaioglu, A. S., Atas, H., Martin, M. J., Cetin-Atalay, R., Atalay, V., & Doğan, T. (2018). Recent applications of deep learning and machine intelligence on in silico drug discovery: methods, tools and databases. *Briefings in Bioinformatics*, *20*(5), 1878–1912. http://dx.doi.org/10.1093/bib/bby061.

Rifaioglu, A. S., Cetin Atalay, R., Cansen Kahraman, D., Doğan, T., Martin, M., & Atalay, V. (2021). MDeePred: novel multi-channel protein featurization for deep learning-based binding affinity prediction in drug discovery. *Bioinformatics*, *37*(5), 693–704. http://dx.doi.org/10.1093/bioinformatics/btaa858.

Rogers, D., & Hahn, M. (2010). Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, *50*(5), 742–754. http://dx.doi.org/10.1021/ci100050t.

Roy, A., Yang, J., & Zhang, Y. (2012). COFACTOR: an accurate comparative algorithm for structure-based protein function annotation. *Nucleic Acids Research*, *40*(W1), W471–W477. http://dx.doi.org/10.1093/nar/gks372.

Schenone, M., Dančík, V., Wagner, B. K., & Clemons, P. A. (2013). Target identification and mechanism of action in chemical biology and drug discovery. *Nature Chemical Biology*, *9*(4), 232–240. http://dx.doi.org/10.1038/nchembio.1199.

Schneider, P., Walters, W. P., Plowright, A. T., Sieroka, N., Listgarten, J., Goodnow, R. A., Fisher, J., Jansen, J. M., Duca, J. S., Rush, T. S., Zentgraf, M., Hill, J. E., Krutoholow, E., Kohler, M., Blaney, J., Funatsu, K., Luebkemann, C., & Schneider, G. (2020). Rethinking drug design in the artificial intelligence era. *Nature Reviews Drug Discovery*, *19*(5), 353–364. http://dx.doi.org/10.1038/s41573-019-0050-3.

Shim, J., Hong, Z.-Y., Sohn, I., & Hwang, C. (2021). Prediction of drug-target binding affinity using similarity-based convolutional neural network. *Scientific Reports*, *11*(1), 4416. http://dx.doi.org/10.1038/s41598-021-83679-y.

Tang, J., Szwajda, A., Shakyawar, S., Xu, T., Hintsanen, P., Wennerberg, K., & Aittokallio, T. (2014). Making sense of large-scale kinase inhibitor bioactivity data sets: A comparative and integrative analysis. *Journal of Chemical Information and Modeling*, *54*(3), 735–743. http://dx.doi.org/10.1021/ci400709d.

Tian, K., Shao, M., Wang, Y., Guan, J., & Zhou, S. (2016). Boosting compound-protein interaction prediction by deep learning. *Methods*, *110*, 64–72. http://dx.doi.org/10.1016/j.ymeth.2016.06.024.

Tsubaki, M., Tomii, K., & Sese, J. (2019). Compound–protein interaction prediction with end-to-end learning of neural networks for graphs and sequences. *Bioinformatics*, *35*(2), 309–318. http://dx.doi.org/10.1093/bioinformatics/bty535.

van Westen, G. J., Swier, R. F., Cortes-Ciriano, I., Wegner, J. K., Overington, J. P., IJzerman, A. P., van Vlijmen, H. W., & Bender, A. (2013). Benchmarking of protein descriptor sets in proteochemometric modeling (part 2): modeling performance of 13 amino acid descriptor sets. *Journal of Cheminformatics*, *5*(1), 42. http://dx.doi.org/10.1186/1758-2946-5-42.

van Westen, G. J., Swier, R. F., Wegner, J. K., IJzerman, A. P., van Vlijmen, H. W., & Bender, A. (2013). Benchmarking of protein descriptor sets in proteochemometric modeling (part 1): comparative study of 13 amino acid descriptor sets. *Journal of Cheminformatics*, *5*(1), 41. http://dx.doi.org/10.1186/1758-2946-5-41.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. arXiv:1706.03762v5.

Wang, F., Wu, F.-X., Li, C.-Z., Jia, C.-Y., Su, S.-W., Hao, G.-F., & Yang, G.-F. (2019). ACID: a free tool for drug repurposing using consensus inverse docking strategy. *Journal of Cheminformatics*, *11*(1), 73. http://dx.doi.org/10.1186/s13321-019-0394-z.

Wang, Y.-B., You, Z.-H., Yang, S., Yi, H.-C., Chen, Z.-H., & Zheng, K. (2020). A deep learning-based method for drug-target interaction prediction based on long short-term memory neural network. *BMC Medical Informatics and Decision Making*, *20*(2), 49. http://dx.doi.org/10.1186/s12911-020-1052-0.

Wang, K., Zhou, R., Li, Y., & Li, M. (2021). DeepDTAF: a deep learning method to predict protein–ligand binding affinity. *Briefings in Bioinformatics*, *22*(5), http://dx.doi.org/10.1093/bib/bbab072, bbab072.

Willighagen, E. L., Mayfield, J. W., Alvarsson, J., Berg, A., Carlsson, L., Jeliazkova, N., Kuhn, S., Pluskal, T., Rojas-Chertó, M., Spjuth, O., Torrance, G., Evelo, C. T., Guha, R., & Steinbeck, C. (2017). The chemistry development kit (CDK) v2.0: atom typing, depiction, molecular formulas, and substructure searching. *Journal of Cheminformatics*, *9*(1), 33. http://dx.doi.org/10.1186/s13321-017-0220-4.

Yang, J., Roy, A., & Zhang, Y. (2013a). BioLiP: a semi-manually curated database for biologically relevant ligand–protein interactions. *Nucleic Acids Research*, *41*(D1), D1096–D1103. http://dx.doi.org/10.1093/nar/gks966.

Yang, J., Roy, A., & Zhang, Y. (2013b). Protein–ligand binding site recognition using complementary binding-specific substructure comparison and sequence profile alignment. *Bioinformatics*, *29*(20), 2588–2595. http://dx.doi.org/10.1093/bioinformatics/btt447.

Yap, C. W. (2011). PaDEL-descriptor: An open source software to calculate molecular descriptors and fingerprints. *Journal of Computational Chemistry*, *32*(7), 1466–1474. http://dx.doi.org/10.1002/jcc.21707.

Zhang, W., Bell, E. W., Yin, M., & Zhang, Y. (2020). Edock: blind protein–ligand docking by replica-exchange monte carlo simulation. *Journal of Cheminformatics*, *12*(1), 37. http://dx.doi.org/10.1186/s13321-020-00440-9.

Zhao, Q., Zhao, H., Zheng, K., & Wang, J. (2022). HyperAttentionDTI: improving drug–protein interaction prediction by sequence-based deep learning with attention mechanism. *Bioinformatics*, *38*(3), 655–662. http://dx.doi.org/10.1093/bioinformatics/btab715.